Business Analytics

Vrije Universiteit Amsterdam - Faculty of Science - Business Analytics - 2025-2026

In today’s data driven world, organizations need expert judgement and sharp analysis to ensure they make the right decisions. Business Analytics enables you to harness the power of data science, big data, statistics and machine learning to optimize results and achieve strategic objectives.

Your ability to combine insights from mathematics, computer science and economics with highly developed quantitative and communication skills will make you key to the success of any organization. The Master’s programme Business Analytics will deepen your knowledge in these areas and give you the opportunity to specialize in Computational Intelligence, Optimization of Business Processes, or Financial Risk Management. The Master’s degree is concluded with a six-month individual internship at a company, which is often the first step to a thriving international career.

Info

Level	Master
Language	English
Duration	2 years
Type	Full Time
Credits	120 EC
Faculty	Faculty of Science

Teaching and examination regulation 2025-2026
More information about this programme

Business Analytics specialisation Computational Intelligence 5
Business Analytics specialisation Computational Intelligence constrained choice 5
Business Analytics specialisation Computational Intelligence compulsory courses 6
Business Analytics Dual study path compulsory courses 6
Business Analytics electives 6
BA Professional study path compulsory courses 7
Business Analytics compulsory courses 7
Business Analytics specialisation Financial Risk Management 8
Business Analytics Dual study path compulsory courses 9
Business Analytics specialisation Financial Risk Management constrained choice 9
Business Analytics specialisation Financial Risk Management compulsory courses 9
Business Analytics electives 9
BA Professional study path compulsory courses 10
Business Analytics compulsory courses 11
Business Analytics specialisation Optimization of Business Processes 11
Business Analytics Dual study path compulsory courses 12
Business Analytics electives 12
Business Analytics specialisation Optimisation of Business Processes constrained choice 13
Business Analytics specialisation Optimisation of Business Processes compulsory courses 13
BA Professional study path compulsory courses 14
Business Analytics compulsory courses 14
Courses 14
Advanced Asset Pricing 14
Advanced Machine Learning 15
Applied Analysis: Financial Mathematics (MSc) 16
Applied Stochastic Modeling 17
Applied Text Mining 1: Methods 18
Continuous Optimization 20
Data Mining Techniques 20
Deep Learning 21
Derivatives 23
Discrete Optimization 24
Dual Work Period 25
Dynamic Programming and Reinforcement Learning 27
Entrepreneurship in Analytics and AI 28
Evolutionary Computing 29
Financial Markets and Institutions 30
Investments 32
Learning Machines 33
Machine Learning for the Quantified Self 34
Master Project Business Analytics 36
Mathematical Optimization 38
Multi-Agent Systems 39
Natural Language Processing 40
Optimization of Business Processes 42
Performance of Networked Systems 42
Programming Large-Scale Parallel Systems 44
Project Optimization of Business Processes 45
Project Reinforcement Learning 46
Quantitative Financial Risk Management 47
Quantitative Investing 48
Queueing Theory 49

Research Seminar Business Analytics 50
Scheduling 50
Statistical Models 51
Stochastic Processes for Finance 52
Supply Chain Lab 53
Time Series Models 54

Business Analytics specialisation Computational Intelligence

Description

Master's Business Analytics is a 2-year programme. The first 4 months are mostly devoted to compulsory courses common to all students in the programme. After that, you decide to continue with a specialisation in

- Computational Intelligence,
- Optimisation of Business Processes,
- or Financial Risk Management.

Another choice you make - this one right at the start - is whether for the first 16 months you

- study full time (Professional study path)
- or combine work and study (Dual study path).

You conclude you Master's degree with a 6-month individual internship at a company (the Master Project).

The specialisation Computational Intelligence prepares you for a role as an academically trained professional specialised in the science to enable a computer to learn a specific task from data. It concentrates on computational methodologies and approaches addressing complex real-world problems to which traditional techniques are inadequate. You will learn advanced methods in data mining, machine learning, and artificial intelligence.

In more detail, your programme has a workload of 120 EC and is structured as follows:

- there are components that are compulsory for all students in the programme (60 EC, see below the tab Business Analytics compulsory courses),
- Dual Work Period is for students in the Dual study path only (12 EC, see the tab Dual study path compulsory courses),
- Project Optimization of Business Processes is compulsory in the Professional study path only, it is an elective in the Dual track (6 EC, see the tab Professional study path compulsory courses),
- there are compulsory courses that are specific to your specialisation (18 EC, see the tab Computational Intelligence compulsory courses),
- there is a constrained choice to make, specific to your specialisation (at least 18 EC for the Professional and at least 12 EC for the Dual study path, see the tab Computational Intelligence constrained choice),
- the remainder (up to 18 EC) you fill with electives from the tab Business Analytics electives.

Additionally, if you combine the Professional study path with a semester abroad,

- the constrained choice requirement is relaxed from at least 18 EC to at least 12 EC,
- instead of up to 18 EC of courses from the tab Business Analytics electives, you complete up to 24 EC of courses while on exchange (advance permission from the Exam Board is required).

Finally,

- note that some of the courses are best followed in a certain order, see the tab Business Analytics electives for recommendations,
- for an alternative/better overview of the programme, see Canvas > Community Master BA > Syllabus > Important Information > Alternative year schedule.

Programme Sections

- Business Analytics specialisation Computational Intelligence constrained choice
- Business Analytics specialisation Computational Intelligence compulsory courses
- Business Analytics Dual study path compulsory courses
- Business Analytics electives
- BA Professional study path compulsory courses
- Business Analytics compulsory courses

Business Analytics specialisation Computational Intelligence constrained choice

Description

Constrained choice for the specialisation Computational Intelligence:

- choose at least 18 EC of courses from this list if you are in the Professional study path,
- choose at least 12 EC if you combine the Professional study path with a semester abroad or if you are in the Dual study path.

Courses

Course Name	Period	Credits	Code
Dynamic Programming and Reinforcement Learning	P2	6EC	XM_0093
Multi-Agent Systems	P2	6EC	XM_0052
Performance of Networked Systems	P2	6EC	X_405105
Learning Machines	P3, P6	6EC	XM_0061
Data Mining Techniques	P5	6EC	X_400108
Machine Learning for the Quantified Self	P6	6EC	XM_40012

Business Analytics specialisation Computational Intelligence compulsory courses

Description

The following courses are compulsory for the specialisation Computational Intelligence, both the Professional and the Dual study path.

Courses

Course Name	Period	Credits	Code
Evolutionary Computing	P1	6EC	X_400111
Deep Learning	P2	6EC	XM_0083
Natural Language Processing	P5	6EC	XM_0121

Business Analytics Dual study path compulsory courses

Description

In the Dual track, you combine work and study during the first 16 months of the programme. The work has to be relevant for the study and this Dual Work Period is granted with 12 EC. Often the external Master Project is carried out at the same organization as the Dual Work Period.

For more information concerning the Dual track, please visithttps://vu-business-analytics.github.io/internship-office/dual_programme.html

Courses

Course Name	Period	Credits	Code
Dual Work Period	Ac. Year (September)	12EC	XM_41010

Business Analytics electives

Description

The credits (up to 18 EC) that remain after the compulsory and constrained-choice components, are to be filled up with electives from this list.

1. Investments is a level-300 course. At most two level-300 courses are allowed in your master programme.
2. Out of the courses Quantitative Investing in period 1 and Advanced Asset Pricing in period 2, at most one is allowed.
3. If you wish to take an elective outside of this list, in particular if you opt for a semester abroad, you must obtain advance permission from the Examination Board.

Furthermore, some of the courses from this list (mind they might also be compulsory/constrained choice courses in your chosen specialisation) are best followed in a certain order:

- If you wish to follow Mathematical Optimization and Supply Chain Lab, it is optimal to do Supply Chain Lab first and Mathematical Optimization next.
- If you wish to follow Mathematical Optimization and Discrete/Continuous Optimization, it is optimal to do Mathematical Optimization first and Discrete/Continuous Optimization next.
- If you wish to follow Project Optimization of Business Processes and the course Optimization of Business

Processes, it is optimal to follow the project first and the course next.
- If you wish to follow Project Reinforcement Learning, it is optimal to do Dynamic Programming and Reinforcement Learning first.
- It is optimal to follow Deep Learning in year 2.

Courses

Course Name	Period	Credits	Code
Programming Large-Scale Parallel Systems	P1	6EC	XM_40017
Quantitative Investing	P1	6EC	E_FIN_QI
Stochastic Processes for Finance	P1+2	6EC	X_400352
Continuous Optimization	P1+2+3	6EC	XMM_400446
Discrete Optimization	P1+2+3	6EC	XMM_400445
Advanced Asset Pricing	P2	6EC	E_FIN_AAP
Deep Learning	P2	6EC	XM_0083
Financial Markets and Institutions	P2	6EC	E_FIN_FMI
Mathematical Optimization	P2	6EC	XM_0051
Performance of Networked Systems	P2	6EC	X_405105
Applied Text Mining 1: Methods	P3	6EC	L_PAMATLW004
Project Optimization of Business Processes	P3	6EC	X_400213
Project Reinforcement Learning	P3	6EC	XM_0120
Learning Machines	P3, P6	6EC	XM_0061
Entrepreneurship in Analytics and AI	P4	6EC	XM_0090
Supply Chain Lab	P4	6EC	E_TSCM_SCL
Time Series Models	P4	6EC	E_EORM_TSM
Applied Analysis: Financial Mathematics (MSc)	P4+5	6EC	XM_0140
Optimization of Business Processes	P4+5	6EC	X_400422
Queueing Theory	P4+5+6	6EC	XMM_400397
Scheduling	P4+5+6	6EC	XMM_400396
Data Mining Techniques	P5	6EC	X_400108
Investments	P5	6EC	E_EBE3_INVES
Natural Language Processing	P5	6EC	XM_0121
Machine Learning for the Quantified Self	P6	6EC	XM_40012

BA Professional study path compulsory courses

Description

Project Optimization of Business Processes is only compulsory in the Professional study path (of either specialisation). In the Dual study path, it is an elective.

Courses

Course Name	Period	Credits	Code
Project Optimization of Business Processes	P3	6EC	X_400213

Business Analytics compulsory courses

Description

The following components are compulsory in either specialisation and in both the Professional and the Dual study path. Everything in this list except for the Master Project must be followed in year 1.

* Before you start your internship for the Master Project, you must have finished the Research Seminar Business Analytics and you must have earned all other programme credits except a shortfall of 6 EC is permissible. If you wish to start on the internship and/or graduation project in January 2026, an additional shortfall of 6 EC is permissible provided it is the credits of a course you have followed in period 2 or period 1 and 2 of 2025-26, have taken part in the examination and are expecting the result to be announced in January 2026.

Courses

Course Name	Period	Credits	Code

Course Name	Period	Credits	Code
Master Project Business Analytics	Ac. Year (September)	36EC	X_400459
Advanced Machine Learning	P1	6EC	XM_0010
Applied Stochastic Modeling	P1+2	6EC	X_400392
Statistical Models	P1+2	6EC	X_400418
Research Seminar Business Analytics	P4+5	6EC	XM_0054

Business Analytics specialisation Financial Risk Management

Description

Master's Business Analytics is a 2-year programme. The first 4 months are mostly devoted to compulsory courses common to all students in the programme. After that, you decide to continue with a specialisation in

- Computational Intelligence,
- Optimisation of Business Processes,
- or Financial Risk Management.

Another choice you make - this one right at the start - is whether for the first 16 months you

- study full time (Professional study path)
- or combine work and study (Dual study path).

You conclude you Master's degree with a 6-month individual internship at a company (the Master Project).

The specialisation Financial Risk Management prepares you for a role as an academically trained professional specialised in using financial instruments to manage costly exposures to risk strategically. It concentrates on principles and techniques of decision making under risk and uncertainty. You will learn advanced methods in a broad field of mathematics and computer science applicable to the financial domain.

In more detail, your programme has a workload of 120 EC and is structured as follows:

- there are components that are compulsory for all students in the programme (60 EC, see below the tab Business Analytics compulsory courses),
- Dual Work Period is for students in the Dual study path only (12 EC, see the tab Dual study path compulsory courses),
- Project Optimization of Business Processes is compulsory in the Professional study path only, it is an elective in the Dual track (6 EC, see the tab Professional study path compulsory courses),
- there are compulsory courses that are specific to your specialisation (18 EC, see the tab Financial Risk Management compulsory courses),
- there is a constrained choice to make, specific to your specialisation (at least 18 EC for the Professional and at least 12 EC for the Dual study path, see the tab Financial Risk Management constrained choice),
- the remainder (up to 18 EC) you fill with electives from the tab Business Analytics electives.

Additionally, if you combine the Professional study path with a semester abroad,

- the constrained choice requirement is relaxed from at least 18 EC to at least 12 EC,
- instead of up to 18 EC of courses from the tab Business Analytics electives, you complete up to 24 EC of courses while on exchange (advance permission from the Exam Board is required).

Finally,

- note that some of the courses are best followed in a certain order, see the tab Business Analytics electives for recommendations,
- for an alternative/better overview of the programme, see Canvas > Community Master BA > Syllabus > Important Information > Alternative year schedule.

Programme Sections

- Business Analytics Dual study path compulsory courses
- Business Analytics specialisation Financial Risk Management constrained choice
- Business Analytics specialisation Financial Risk Management compulsory courses
- Business Analytics electives
- BA Professional study path compulsory courses

- Business Analytics compulsory courses

Business Analytics Dual study path compulsory courses

Description

In the Dual track, you combine work and study during the first 16 months of the programme. The work has to be relevant for the study and this Dual Work Period is granted with 12 EC. Often the external Master Project is carried out at the same organization as the Dual Work Period.

For more information concerning the Dual track, please visithttps://vu-business-analytics.github.io/internship-office/dual_programme.html

Courses

Course Name	Period	Credits	Code
Dual Work Period	Ac. Year (September)	12EC	XM_41010

Business Analytics specialisation Financial Risk Management constrained choice

Description

Constrained choice for the specialisation Financial Risk Management:

- choose at least 18 EC of courses from this list if you are in the Professional study path,
- choose at least 12 EC if you combine the Professional study path with a semester abroad or if you are in the Dual study path.

* Out of the courses Quantitative Investing in period 1 and Advanced Asset Pricing in period 2, at most one is allowed.

Courses

Course Name	Period	Credits	Code
Quantitative Investing	P1	6EC	E_FIN_QI
Advanced Asset Pricing	P2	6EC	E_FIN_AAP
Dynamic Programming and Reinforcement Learning	P2	6EC	XM_0093
Financial Markets and Institutions	P2	6EC	E_FIN_FMI
Mathematical Optimization	P2	6EC	XM_0051
Derivatives	P4	6EC	E_FIN_DER
Data Mining Techniques	P5	6EC	X_400108
Quantitative Financial Risk Management	P5	6EC	E_FIN_QFRM

Business Analytics specialisation Financial Risk Management compulsory courses

Description

The following courses are compulsory for the specialisation Financial Risk Management, both the Professional and the Dual study path.

* Investments is of level 300. At most two 300-level courses are allowed in your master programme.

Courses

Course Name	Period	Credits	Code
Stochastic Processes for Finance	P1+2	6EC	X_400352
Applied Analysis: Financial Mathematics (MSc)	P4+5	6EC	XM_0140
Investments	P5	6EC	E_EBE3_INVES

Business Analytics electives

Description

The credits (up to 18 EC) that remain after the compulsory and constrained-choice components, are to be filled up with electives from this list.

1. Investments is a level-300 course. At most two level-300 courses are allowed in your master programme.
2. Out of the courses Quantitative Investing in period 1 and Advanced Asset Pricing in period 2, at most one is allowed.
3. If you wish to take an elective outside of this list, in particular if you opt for a semester abroad, you must obtain advance permission from the Examination Board.

Furthermore, some of the courses from this list (mind they might also be compulsory/constrained choice courses in your chosen specialisation) are best followed in a certain order:

- If you wish to follow Mathematical Optimization and Supply Chain Lab, it is optimal to do Supply Chain Lab first and Mathematical Optimization next.
- If you wish to follow Mathematical Optimization and Discrete/Continuous Optimization, it is optimal to do Mathematical Optimization first and Discrete/Continuous Optimization next.
- If you wish to follow Project Optimization of Business Processes and the course Optimization of Business Processes, it is optimal to follow the project first and the course next.
- If you wish to follow Project Reinforcement Learning, it is optimal to do Dynamic Programming and Reinforcement Learning first.
- It is optimal to follow Deep Learning in year 2.

Courses

Course Name	Period	Credits	Code
Programming Large-Scale Parallel Systems	P1	6EC	XM_40017
Quantitative Investing	P1	6EC	E_FIN_QI
Stochastic Processes for Finance	P1+2	6EC	X_400352
Continuous Optimization	P1+2+3	6EC	XMM_400446
Discrete Optimization	P1+2+3	6EC	XMM_400445
Advanced Asset Pricing	P2	6EC	E_FIN_AAP
Deep Learning	P2	6EC	XM_0083
Financial Markets and Institutions	P2	6EC	E_FIN_FMI
Mathematical Optimization	P2	6EC	XM_0051
Performance of Networked Systems	P2	6EC	X_405105
Applied Text Mining 1: Methods	P3	6EC	L_PAMATLW004
Project Optimization of Business Processes	P3	6EC	X_400213
Project Reinforcement Learning	P3	6EC	XM_0120
Learning Machines	P3, P6	6EC	XM_0061
Entrepreneurship in Analytics and AI	P4	6EC	XM_0090
Supply Chain Lab	P4	6EC	E_TSCM_SCL
Time Series Models	P4	6EC	E_EORM_TSM
Applied Analysis: Financial Mathematics (MSc)	P4+5	6EC	XM_0140
Optimization of Business Processes	P4+5	6EC	X_400422
Queueing Theory	P4+5+6	6EC	XMM_400397
Scheduling	P4+5+6	6EC	XMM_400396
Data Mining Techniques	P5	6EC	X_400108
Investments	P5	6EC	E_EBE3_INVES
Natural Language Processing	P5	6EC	XM_0121
Machine Learning for the Quantified Self	P6	6EC	XM_40012

BA Professional study path compulsory courses

Description

Project Optimization of Business Processes is only compulsory in the Professional study path (of either specialisation). In the Dual study path, it is an elective.

Courses

Course Name	Period	Credits	Code
Project Optimization of Business Processes	P3	6EC	X_400213

Business Analytics compulsory courses

Description

The following components are compulsory in either specialisation and in both the Professional and the Dual study path. Everything in this list except for the Master Project must be followed in year 1.

* Before you start your internship for the Master Project, you must have finished the Research Seminar Business Analytics and you must have earned all other programme credits except a shortfall of 6 EC is permissible. If you wish to start on the internship and/or graduation project in January 2026, an additional shortfall of 6 EC is permissible provided it is the credits of a course you have followed in period 2 or period 1 and 2 of 2025-26, have taken part in the examination and are expecting the result to be announced in January 2026.

Courses

Course Name	Period	Credits	Code
Master Project Business Analytics	Ac. Year (September)	36EC	X_400459
Advanced Machine Learning	P1	6EC	XM_0010
Applied Stochastic Modeling	P1+2	6EC	X_400392
Statistical Models	P1+2	6EC	X_400418
Research Seminar Business Analytics	P4+5	6EC	XM_0054

Business Analytics specialisation Optimization of Business Processes

Description

Master's Business Analytics is a 2-year programme. The first 4 months are mostly devoted to compulsory courses common to all students in the programme. After that, you decide to continue with a specialisation in

- Computational Intelligence,
- Optimisation of Business Processes,
- or Financial Risk Management.

Another choice you make - this one right at the start - is whether for the first 16 months you

- study full time (Professional study path)
- or combine work and study (Dual study path).

You conclude you Master's degree with a 6-month individual internship at a company (the Master Project).

The specialisation Optimisation of Business Processes prepares you for a role as an academically trained professional who tackles quantitative business problems with the aid of mathematical algorithms for decision making. It concentrates on improving business processes through data analysis, predictive modelling, and optimisation. You will learn advanced methods from data science, machine learning, machine learning, and operations research.

In more detail, your programme has a workload of 120 EC and is structured as follows:

- there are components that are compulsory for all students in the programme (60 EC, see below the tab Business Analytics compulsory courses),
- Dual Work Period is for students in the Dual study path only (12 EC, see the tab Dual study path compulsory courses),
- Project Optimization of Business Processes is compulsory in the Professional study path only, it is an elective in the Dual track (6 EC, see the tab Professional study path compulsory courses),
- there are compulsory courses that are specific to your specialisation (18 EC, see the tab Optimisation of Business Processes compulsory courses),
- there is a constrained choice to make, specific to your specialisation (at least 18 EC for the Professional and at least 12 EC for the Dual study path, see the tab Optimisation of Business Processes constrained choice),
- the remainder (up to 18 EC) you fill with electives from the tab Business Analytics electives.

Additionally, if you combine the Professional study path with a semester abroad,

- the constrained choice requirement is relaxed from at least 18 EC to at least 12 EC,
- instead of up to 18 EC of courses from the tab Business Analytics electives, you complete up to 24 EC of courses while on exchange (advance permission from the Exam Board is required).

Finally,

- note that some of the courses are best followed in a certain order, see the tab Business Analytics electives for recommendations,
- for an alternative/better overview of the programme, see Canvas > Community Master BA > Syllabus > Important Information > Alternative year schedule.

Programme Sections

- Business Analytics Dual study path compulsory courses
- Business Analytics electives
- Business Analytics specialisation Optimisation of Business Processes constrained choice
- Business Analytics specialisation Optimisation of Business Processes compulsory courses
- BA Professional study path compulsory courses
- Business Analytics compulsory courses

Business Analytics Dual study path compulsory courses

Description

In the Dual track, you combine work and study during the first 16 months of the programme. The work has to be relevant for the study and this Dual Work Period is granted with 12 EC. Often the external Master Project is carried out at the same organization as the Dual Work Period.

For more information concerning the Dual track, please visithttps://vu-business-analytics.github.io/internship-office/dual_programme.html

Courses

Course Name	Period	Credits	Code
Dual Work Period	Ac. Year (September)	12EC	XM_41010

Business Analytics electives

Description

The credits (up to 18 EC) that remain after the compulsory and constrained-choice components, are to be filled up with electives from this list.

1. Investments is a level-300 course. At most two level-300 courses are allowed in your master programme.
2. Out of the courses Quantitative Investing in period 1 and Advanced Asset Pricing in period 2, at most one is allowed.
3. If you wish to take an elective outside of this list, in particular if you opt for a semester abroad, you must obtain advance permission from the Examination Board.

Furthermore, some of the courses from this list (mind they might also be compulsory/constrained choice courses in your chosen specialisation) are best followed in a certain order:

- If you wish to follow Mathematical Optimization and Supply Chain Lab, it is optimal to do Supply Chain Lab first and Mathematical Optimization next.
- If you wish to follow Mathematical Optimization and Discrete/Continuous Optimization, it is optimal to do Mathematical Optimization first and Discrete/Continuous Optimization next.
- If you wish to follow Project Optimization of Business Processes and the course Optimization of Business Processes, it is optimal to follow the project first and the course next.
- If you wish to follow Project Reinforcement Learning, it is optimal to do Dynamic Programming and Reinforcement Learning first.
- It is optimal to follow Deep Learning in year 2.

Courses

Course Name	Period	Credits	Code
Programming Large-Scale Parallel Systems	P1	6EC	XM_40017
Quantitative Investing	P1	6EC	E_FIN_QI
Stochastic Processes for Finance	P1+2	6EC	X_400352
Continuous Optimization	P1+2+3	6EC	XMM_400446
Discrete Optimization	P1+2+3	6EC	XMM_400445
Advanced Asset Pricing	P2	6EC	E_FIN_AAP
Deep Learning	P2	6EC	XM_0083
Financial Markets and Institutions	P2	6EC	E_FIN_FMI
Mathematical Optimization	P2	6EC	XM_0051
Performance of Networked Systems	P2	6EC	X_405105
Applied Text Mining 1: Methods	P3	6EC	L_PAMATLW004
Project Optimization of Business Processes	P3	6EC	X_400213
Project Reinforcement Learning	P3	6EC	XM_0120
Learning Machines	P3, P6	6EC	XM_0061
Entrepreneurship in Analytics and AI	P4	6EC	XM_0090
Supply Chain Lab	P4	6EC	E_TSCM_SCL
Time Series Models	P4	6EC	E_EORM_TSM
Applied Analysis: Financial Mathematics (MSc)	P4+5	6EC	XM_0140
Optimization of Business Processes	P4+5	6EC	X_400422
Queueing Theory	P4+5+6	6EC	XMM_400397
Scheduling	P4+5+6	6EC	XMM_400396
Data Mining Techniques	P5	6EC	X_400108
Investments	P5	6EC	E_EBE3_INVES
Natural Language Processing	P5	6EC	XM_0121
Machine Learning for the Quantified Self	P6	6EC	XM_40012

Business Analytics specialisation Optimisation of Business Processes constrained choice

Description

Constrained choice for the specialisation Optimization of Business Processes:

- choose at least 18 EC of courses from this list if you are in the Professional study path,
- choose at least 12 EC if you combine the Professional study path with a semester abroad or if you are in the Dual study path.

Courses

Course Name	Period	Credits	Code
Evolutionary Computing	P1	6EC	X_400111
Continuous Optimization	P1+2+3	6EC	XMM_400446
Discrete Optimization	P1+2+3	6EC	XMM_400445
Performance of Networked Systems	P2	6EC	X_405105
Entrepreneurship in Analytics and AI	P4	6EC	XM_0090
Supply Chain Lab	P4	6EC	E_TSCM_SCL
Applied Analysis: Financial Mathematics (MSc)	P4+5	6EC	XM_0140
Scheduling	P4+5+6	6EC	XMM_400396
Data Mining Techniques	P5	6EC	X_400108

Business Analytics specialisation Optimisation of Business Processes compulsory courses

Description

The following courses are compulsory for the specialisation Optimisation of Business Processes, both the Professional and the Dual study path.

Courses

Course Name	Period	Credits	Code
Dynamic Programming and Reinforcement Learning	P2	6EC	XM_0093

Course Name	Period	Credits	Code
Mathematical Optimization	P2	6EC	XM_0051
Optimization of Business Processes	P4+5	6EC	X_400422

BA Professional study path compulsory courses

Description

Project Optimization of Business Processes is only compulsory in the Professional study path (of either specialisation). In the Dual study path, it is an elective.

Courses

Course Name	Period	Credits	Code
Project Optimization of Business Processes	P3	6EC	X_400213

Business Analytics compulsory courses

Description

The following components are compulsory in either specialisation and in both the Professional and the Dual study path. Everything in this list except for the Master Project must be followed in year 1.

* Before you start your internship for the Master Project, you must have finished the Research Seminar Business Analytics and you must have earned all other programme credits except a shortfall of 6 EC is permissible. If you wish to start on the internship and/or graduation project in January 2026, an additional shortfall of 6 EC is permissible provided it is the credits of a course you have followed in period 2 or period 1 and 2 of 2025-26, have taken part in the examination and are expecting the result to be announced in January 2026.

Courses

Course Name	Period	Credits	Code
Master Project Business Analytics	Ac. Year (September)	36EC	X_400459
Advanced Machine Learning	P1	6EC	XM_0010
Applied Stochastic Modeling	P1+2	6EC	X_400392
Statistical Models	P1+2	6EC	X_400418
Research Seminar Business Analytics	P4+5	6EC	XM_0054

Courses

Advanced Asset Pricing

Course Code	E_FIN_AAP
Credits	6
Period	P2
Course Level	400
Language Of Tuition	English
Faculty	School of Business and Economics
Course Coordinator	A. Pitkäjärvi
Examiner	A. Pitkäjärvi
Teaching Staff	M.A. de Vries, A. Pitkäjärvi
Teaching method(s)	Study Group, Lecture

Course Objective

This course aims to deepen your knowledge in the field of asset pricing and investments.

After completing this course, you should:

- Understand the determinants of equity returns.
- Be familiar with the traditional and behavioral approaches to asset pricing.
- Appreciate how various market frictions and constraints affect asset prices.
- Acquire an academic and critical attitude towards competing theories in the field of asset pricing and investments.
- Be able to perform advanced analyses on large data sets using software such as Microsoft Excel.

Course Content

Starting from basic (undergraduate) investments knowledge, this course centers around the issues of asset pricing and investments. In the first week we revisit the well-known mean-variance framework and derive the standard CAPM. Starting from the second week, we carefully study the assumptions underlying the CAPM framework and ask ourselves what they imply for asset pricing. Examples include the assumption of mean-variance utility, rational expectations, and complete arbitrage. Throughout the course, neoclassical and behavioral theories are confronted with each other. The course includes a combination of theory and empirics. Students will compete in a trading game in which they can directly apply their newly acquired knowledge and experience the real-life issues associated with investing.

Additional Information Teaching Methods

The course features lectures, knowledge clips, and tutorials. The focus of the assignments is to apply the theoretical knowledge from the lectures and knowledge clips to real world problems using actual stock market data. In addition to gaining a deeper understanding of the topics in the course, the assignments will train you in quantitative skills you will need later in your career, and prepare you for your thesis and for similar assignments in other courses.

Method of Assessment

Assignments and written exam.

Entry Requirements

You should be familiar with investments at the level of Bodie, Kane & Marcus, Investments. Undergraduate level knowledge of statistics and mathematics is also required (e.g., Berenson, Levine, Krehbiel: Basic Business Statistics; Sydsaeter and Hammond: Essential Mathematics for Economic Analysis; Sydsaeter, Hammond, Seierstad, and Strom: Further Mathematics for Economic Analysis).

Literature

Selected research articles.

Additional Information Target Audience

You are not allowed to take this course if you have already taken Quantitative Investing. This course can have an in-depth follow-up by choosing the investments learning line, consisting of an appropriate investments related Research Project in Period 3, as well as related electives in Periods 4 and 5 (e.g., Derivatives, Institutional Investments and Asset Liability Management, Macro and International Finance, Quantitative Financial Risk Management).

Recommended background knowledge

You are expected to be able to work with data and perform analyses using software such as Microsoft Excel, or a programming language such as R or Python.

Advanced Machine Learning

Course Code	XM_0010
Credits	6
Period	P1
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. S. Bhulai
Examiner	prof. dr. S. Bhulai
Teaching Staff	prof. dr. S. Bhulai

Teaching method(s)	Lecture

Course Objective

There are several learning objectives for this course. After completion of this course, the student should be able to:

- understand the capabilities and the limitations of machine learning,
- implement machine learning algorithms in Python,
- know relevant machine learning algorithms for both supervised and unsupervised learning problems,
- select the right machine learning models for real-world use cases,
- understand when to apply online learning, reinforcement learning, and deep learning,
- interpret the outcomes of machine learning algorithms.

Course Content

Machine learning is the science of getting computers to act without being explicitly programmed. It is so pervasive today that people use it in everyday life without knowing it. In this course, you will learn about the most effective machine learning techniques and gain practice implementing them and getting them to work yourself. We will discuss the theoretical underpinnings as well as the practical know-how needed to apply these techniques to new problems.

Additional Information Teaching Methods

Lectures (14 x 2 hours), including guest speakers.

Method of Assessment

Written exam.

Literature

Slides and additional material that will be posted on Canvas.

Additional Information Target Audience

mBA, mBA-D, mMath, mCS

Recommended background knowledge

The VU course Linear Algebra and the VU course Statistics, or equivalent courses, are preferred.

Applied Analysis: Financial Mathematics (MSc)

Course Code	XM_0140
Credits	6
Period	P4+5
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. A.C.M. Ran
Examiner	prof. dr. A.C.M. Ran
Teaching Staff	prof. dr. A.C.M. Ran
Teaching method(s)	Seminar, Written partial exam, Lecture

Course Objective

This course is taught jointly with X_400076 Applied Analysis: Financial Mathematics for BSc Mathematics, see the studyguide of X_400076.

Course Content

See the studyguide of X_400076

Additional Information Teaching Methods

See the studyguide of X_400076

Method of Assessment

See the studyguide of X_400076

Literature

See the studyguide of X_400076

Additional Information Target Audience

MSc Business Analytics, MSc Mathematics

Note that this course is not intended for students who followed X_400076 in their Bachelor's, due to considerable overlap.

Recommended background knowledge

See the studyguide of X_400076

Applied Stochastic Modeling

Course Code	X_400392
Credits	6
Period	P1+2
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. R. Bekker
Examiner	dr. R. Bekker
Teaching Staff	dr. R. Bekker
Teaching method(s)	Seminar, Lecture

Course Objective

During this course you will get acquainted with the most often used stochastic models and how they are applied in practice. The emphasis is on the variety of stochastic models (and their analysis) that appear in practice, rather than an in-depth study of a single-class of models. During the course you learn to handle such practically motivated problems as an independent researcher; this means that you:

- are able to determine the appropriate model
- can formulate the problem mathematically correct
- are able to solve the stochastic model
- know how to interpret the outcome.

Course Content

This course deals with a number of stochastic modeling techniques that are often used in practice. They are motivated by showing the business context in which they are used. Topics we deal with are: time-dependent Poisson processes and infinite-server queues, renewal processes and simulation, birth-death-processes, basic queueing models, and inventory models. We also repeat and extend certain parts of probability theory.

Additional Information Teaching Methods

Lecture and instruction.

Method of Assessment

Written examination and two hand-in assignments (one in each period).
The assignment in period 1 is graded as a pass-fail, and the assignment in period 2 counts for 20%; the written examination counts for 80%. The weighted average grade should be sufficient, but for the written examination there

is a minimum requirement of a 5.0. The same grading scheme applies to the resit, unless the grade for the resit exam is higher than the grade for the handin-assignment in period 2, in which case the grade for the resit is determined entirely by the resit exam.
There is no resit for the hand-in assignment in period 2. The hand-in assignment in period 1 may be repaired in case of a fail.

Literature

- Lecture notes of Ger Koole (link will be made available via Canvas).
- Recommended: H.C. Tijms, A First Course in Stochastic Models, 2003. This is available as e-book via the VU library (ubvu), free of charge.
- Two scientific paper(s) will be made available via Canvas.

Additional Information Target Audience

mBA, mMath

Recommended background knowledge

Probability theory, Poisson process, Markov chains in continuous time

Applied Text Mining 1: Methods

Course Code	L_PAMATLW004
Credits	6
Period	P3
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Humanities
Course Coordinator	dr. E. Maks
Examiner	dr. E. Maks
Teaching Staff	dr. E. Maks, dr. L.G. de Passos Morgado da Costa BA
Teaching method(s)	Seminar

Course Objective

In this short and intensive course, students will gain hands-on experience conducting a machine learning experiment to solve an NLP task in a group work setting.

At the end of this course students will be able to independently:

- Understand the specific needs of NLP tasks starting from literature review
- Preprocess and analyse existing NLP datasets and their annotations
- Design, set up and run machine learning experiments (developing both feature-based + transformer-based models) to solve NLP tasks
- Evaluate and interpret the results of NLP experiments
- Design and implement linguistically-informed error analyses for NLP tasks

Course Content

In this course students will work closely as a group to conduct an NLP experiment. This is a full-time course, where students are expected to commit ~40h/week and progressively work towards the completion of a series of NLP experiments. Each week, the different phases of the experiment will be covered (with deliverables).

Please note that the course is designed to be a practical course, and while some content will be reviewed, the students are expected to already have the necessary background to independently (as a group) conduct the NLP experiment.

During this course:

- Students will work on understanding the task definition and formulating an experiment design.
- Students will also work with existing datasets, understand how the data was annotated, and the need to preprocess the data and format it in a way that it can be used for the experiment;
- Students will work on building supervised classifiers for the experiment at hand. They follow standard experimental protocols.
- Students will be guided through reporting essential information in an NLP experiment, including: data

preprocessing and statistics, general architecture of classifiers, features (if applicable), as well as evaluation and its metrics.
- These experimental protocols will be relevant for both traditional and transformer-based classifiers and both will be part of the course.
- Students will work on designing and implementing an error analysis for the task at hand. Students will need to understand the linguistic dimensions of the task and system's performance. Students will analyse potential sources of error and design tests to verify their hypotheses

Additional Information Teaching Methods

This course is mostly practical. The lectures will review the basic theory related to each methodological step, which the students will apply to solve the task. A few sessions will be devoted to theoretical aspects, while other sessions will be interactive and devoted to discuss progress, doubts and problems encountered. Students will work in groups.

Students are expected to commit 40 hours per week (including both lectures and group work). Attendance is mandatory for the two weekly lectures/practical sessions.

Contact hours: 2x2 hours per week

Method of Assessment

The course includes a group+individual assignment component (with weekly deliverables) and an individual exam component consisting of a TestVision exam and, potentially, an oral exam related to the weekly assignments. All graded components must be graded at least 5.5.

Entry Requirements

This course requires programming experience and the ability to independently implement and understand supervised machine learning systems (including: feature-based models such as Logistic Regression and SVMs; deep neural methods such as Bi-LSTMs; and transformer-based models such as fine-tuned BERT models).

In addition, this course also assumes prior knowledge of core linguistic concepts in morphology, syntax and semantics.

Students of the Language and AI/HLT programs who have successfully completed all prior courses fulfill all requirements.

Other students should ensure they have had one course on Machine Learning (e.g., Machine Learning for NLP, Machine Learning for the Quantified Self, Machine Learning). In addition, they should also have at least one NLP course where the foundational linguistic concepts mentioned above were covered (e.g., Introduction to Human Language Technology, Natural Language Processing Technology, Language as Data, Subjectivity Mining).

Literature

The literature of the course is tied to the NLP task chosen for each run of the course. Literature will mostly comprise academic papers describing a specific NLP task, and academic papers proposing systems to address that same task.

Additional Information Target Audience

The target audience of this course is mainly students in the Language and AI track of the MA in Linguistics. Other students are welcome insofar they meet the entry requirements.

Additional Information

This is a concise course requiring 40 hours of weekly commitment. Assignments are mostly group assignments and students need to contribute to these assignments equally.

Students who are not able to commit the required time should not register for this course.

Explanation Canvas

Course materials will be made available on Canvas.

Recommended background knowledge

See Entry requirements.

Continuous Optimization

Course Code	XMM_400446
Credits	6
Period	P1+2+3
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. S. Bhulai
Examiner	
Teaching Staff	
Teaching method(s)	

Course Objective

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Course Content

This course is part of the joint national master programme in mathematics. For schedules, course locations and course descriptions see https://elo.mastermath.nl.

Additional Information Teaching Methods

-

Method of Assessment

-

Additional Information Target Audience

mMath; mBA

Custom Course Registration

You have to register your participation in each Mastermath course via https://elo.mastermath.nl/login/. Registration is mandatory and absolutely necessary for transferring your grades from Mastermath to the administration of your university.

Data Mining Techniques

Course Code	X_400108
Credits	6
Period	P5
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. M. Hoogendoorn
Examiner	prof. dr. M. Hoogendoorn
Teaching Staff	dr. S. Yu, prof. dr. M. Hoogendoorn, B. Borsboom
Teaching method(s)	Computer lab, Lecture

Course Objective

The aim of the course is that students acquire data mining knowledge and skills that they can apply in a business environment. More precisely, the following learning goals are distinguished:

1. Understand the data mining process
2. Understand exploratory data analysis
3. Understand feature engineering

4. Understand classification techniques
5. Understand regression techniques
6. Understand association rules and recommender systems
7. Understand big data engineering and big data infrastructures
8. Understand current state of the art in data mining
9. Understand ethical concerns and evaluation of results
10. Learning to argue for choices (rationale)

The course if focused on the Dublin descriptors Applying Knowledge and Understanding (since the aim is to improve the practical skills); Making judgements (which technique is appropriate, how to best apply Data Mining for a specific case); Communication skills (how to report on your approach, choices and your results), and Learning skills (able to find new relevant techniques, assess their suitability, etc.).

How the aims are to be achieved: Students will acquire knowledge and skills mainly through the following: an overview of the most common data mining algorithms and techniques (in lectures), a survey of typical and interesting data mining applications, and practical assignments to gain "hands on" experience. The application of skills in a business environment will be simulated through various assignments of the course.

Course Content

The course is intended to introduce Data Mining Techniques to students that are new to the field as well as to more experienced students. The main aim is to gain a more practical perspective towards Data Mining Techniques/Machine Learning. Lectures will cover more basic things for those new to the field (general introduction into Data Mining, classical algorithms such as decision trees, association rules, neural networks, ensemble learning, etc.) and on top will discuss advanced topics including deep learning, recommender systems, big data infrastructures, and text mining. A number of successful applications in the area will also be discussed. In addition to lectures, there will be an extensive practical part, where students will experiment with various data mining algorithms and data sets. The grade for the course will be based on these practical assignments.

Additional Information Teaching Methods

Lectures (h) and practical sessions (pra). Lectures are planned to be interactive: there will be small questions, etc.

Method of Assessment

Practical assignments. There will be two assignments done in groups of three. For the first assignment there is a choice: going for a basic version of the assignment (suited for those new to the domain) or a more advanced one (for students with more experience). The second assignment is the same for all and will involve an in-class Kaggle competition. The first assignment counts for 40% and the second for 60%.

An interview will take place on an individual student level to make sure that (1) each student understands the content of the graded material, and (2) has made a sufficient contribution to the group work. Only when this check is passed will a final grade be given. Furthermore, the grade of both assignments needs to be sufficient to pass the course. Finally, attendance of the final lecture (in which the final ranking of the Kaggle competition is announced and selected groups of students will be asked to present their solution) is mandatory as it contributes to the learning objectives of the course.

There is an option to resit one of the assignments when the assignment has been submitted but an insufficient grade is obtained (only when the other assignment grade is sufficient). This involves improving your assignment to at least a sufficient level within two weeks after having received your grade for the second assignment. The maximum grade of the resit is a 6.0.

Literature

Optional: Ian H. Witten, Eibe Frank, Mark A. Hall, and Christopher J. Pal. Data Mining: Practical Machine Learning Tools and Techniques (Fourth Edition). Morgan Kaufmann, 2016 978-0128042915.

Additional Information Target Audience

Master Artificial Intelligence, Master Computer Science, Master Bioinformatics and Systems Biology, Master Business Analytics

Recommended background knowledge

1. Statistics
2. Programming Skills

Deep Learning

Course Code	XM_0083
Credits	6
Period	P2
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. S. Yu
Examiner	dr. S. Yu
Teaching Staff	dr. S. Yu, dr. M. Cochez, dr. P. Bloem
Teaching method(s)	Practical, Lecture

Course Objective

1. Knowledge and understanding

- Be able to explain the main components of a deep learning architecture: fully-connected layers, convolutional layers, activation functions, LSTM, and GRU.
- Be able to explain and derive the backpropagation algorithm.
- Be able to explain different commonly used deep architectures.
- Be able to explain Variational Autoencoders and unsupervised representation learning.
- Be able to explain the difference between discriminative and generative models.

2. Applying knowledge and understanding

- How before-mentioned layers work and where to use them.
- How to formulate a neural network for a specific problem.
- How to analyze the performance of a neural network.
- What neural network fits best for a given problem.

3. Making judgments

- What deep learning model to use for a given problem (e.g., generative vs discriminative).
- What layers of a neural network are suitable for a given problem.

4. Communication skills

- Presenting an analysis in a written form (a short report) for each assignment.

5. Learning skills

- Able to read (some) state-of-the-art papers.
- Able to develop (to some extent) available deep learning libraries.

Course Content

Deep learning becomes the leading learning and modeling paradigm in machine learning. During this course, we will present basic components of deep learning, such as:

- different layers (e.g., linear layers, convolutional layers, pooling layers, recurrent layers);
- non-linear activation functions (e.g., sigmoid, ReLU);
- backpropagation;
- learning algorithms (e.g., ADAM);
- other (e.g., dropout).

Further, we will show how to build deep architectures like AlexNet and ResNet. We will explain potential pitfalls and possible solutions, e.g., by using residual connections and dense architectures.

After discussing discriminative models, we will turn them into generative models. We will start with linear latent variable models like the probabilistic PCA. Then we will discuss a non-linear version of the pPCA, namely, Variational Auto-Encoders (VAEs). Both pPCA and VAE are so-called prescribed models that require formulating the likelihood function. Next, we will introduce unsupervised representation learning, different ways to improve VAEs, and how to obtain disentangled representations.

At the end of the course, we will outline recent developments in deep learning. Namely, we will discuss Reinforcement Learning and Deep Reinforcement Learning. In the end, we will introduce generalization and explainabilility - two fundamental topics of deep learning.

Additional Information Teaching Methods

The course consists of two parts: a digital exam and practical assignments. The digital exam is supported by lectures (two/three per week). The assignments are supported by practical sessions led by TAs.

The first two assignments are carried out individually and the remaining assignments are done in small groups. No resit is possible for the practical assignments.

Method of Assessment

Final exam (50%) and practical assignments (50%). There are 4 assignments (2 of them are done individually; 2 of them are done in groups).

The final exam must be passed with a sufficient grade (equivalent to a grade of 5.5 or higher).

There is a resit for the exam. The practical assignments cannot be redone.

Literature

The literature will be made available on Canvas.
Two suggested books:

- Goodfellow, I., Bengio, Y., & Courville, A. (2016). "Deep learning". MIT press.
- Tomczak, J. M. (2022). "Deep Generative Modeling". Springer, Cham.

Additional Information Target Audience

Master Artificial Intelligence
Master Business Analytics

Recommended background knowledge

- Calculus
- Linear Algebra
- Statistics & Probability Theory
- Programming (Python)
- Machine Learning

Derivatives

Course Code	E_FIN_DER
Credits	6
Period	P4
Course Level	400
Language Of Tuition	English
Faculty	School of Business and Economics
Course Coordinator	dr. N.J. Seeger
Examiner	dr. N.J. Seeger
Teaching Staff	dr. N.J. Seeger
Teaching method(s)	Computer lab, Study Group, Lecture

Course Objective

The primary objective of this course is to provide students with an advanced introduction to derivative instruments. By the end of the course students should have a sound understanding of pricing concepts, practical applicability, risk management concepts related to derivatives and implementing such concepts in a programming language.

Course Content

In todays financial world, the role of derivatives gets increasingly important. Banks and pension funds use derivatives to manage their balance sheet risk, corporate treasuries need derivatives for mitigation of international trade risk, insurance companies actively apply derivatives strategically in order to hedge long term interest rate exposures. Worldwide derivatives trading has exploded to unprecedented levels in the last decades. Therefore, a sound understanding of derivatives is indispensable for anyone pursuing a job in finance.

The course aims to help students in developing a general understanding of the fundamental principles related to derivative instruments. When we try to understand derivative instruments we will ask questions like:

1. How do derivative instruments work?
2. Is it possible to decompose derivatives in basic assets?
3. How to determine the fair value of derivative instruments?
4. What are the risks of using derivative instruments?
5. How are derivative instruments applied in practice and are there any relevant operational issues in the real world?

Hence, the course focuses on facilitating conceptual understanding of derivative instruments and of the methods that are needed to apply derivatives in different settings of finance applications; whether it is for trading purposes, structuring products, risk management, etc.

Applying methods such as e.g. pricing or hedging derivatives to practical applications knowledge of programming is absolutely necessary.
In separate tutorial sessions we will implement theoretical concepts discussed in class using in a programming language to make them available for finance applications.

The field of derivatives is one of the most mathematically sophisticated in finance. Therefore, to understand derivatives it is inevitable to deal with mathematical methods. However, we want to emphasize that in the course mathematical methods are primarily used as tools to understand derivatives. We intend to serve a balanced mix of theory, intuition and practical aspects.

The course will treat the following subjects:

- Why derivatives?
- Forwards, futures and options
- Pricing concepts of derivative instruments
- Discrete and continuous time option pricing models
- Understanding Black-Scholes formula
- Beyond Black-Scholes (stochastic volatility and jumps)
- Hedging strategies

Additional Information Teaching Methods

The course spans a period of six weeks. There will be a mixture of content driven lecture type sessions and Q&A tutorial like sessions, in which the course material is presented and practiced.

Method of Assessment

The final grade of the course is achieved by solving course-content based problems in final exam.

Entry Requirements

Students entering this course should be familiar with the basic corporate finance principles and techniques (e.g. Berk/DeMarzo, Corporate Finance) and investment management concepts (e.g. Bodie et al, Investments). It is important to notice that this course is not an introductory course to derivatives, but rather an advanced introduction. To follow the course students need to have understanding of the basic concepts of derivatives already. In order to follow the course material right from the start it is recommended to review the derivatives material that has been covered in the bachelor finance courses. Furthermore, before the course starts a slides set will be made available that consists basic material. This material is the starting point of the lecture and students are required to have mastered such material to be able to actively use in during the course at all times.

Literature

- Lecture slides
- John Hull: Options, Futures and other Derivatives, 11th Edition, 2021

Further References:

- Das, R.K. and S.R. Sundaram: Dervatives: Principles and Practice, McGRAW-Hill International Edition, 2nd ed., 2017
- Jarrow, R. and A. Chatterjea: An Introduction to Derivative Securities, Financial Markets, and Risk Management, W. W. Norton & Company, 2013
- Baxter/Rennie: Financial Calculus, Cambridge, 1996. - Neftci: Principles of Financial Engineering, Elsevier, 2nd edition, 2008.
- Bingham/Kiesel: Risk-Neutral Valuation: Pricing and Hedging of Financial Derivatives, Springer, 2004.
- Björk, T.: Arbitrage Theory in Continuous Time, Oxford University Press, 2004.

Discrete Optimization

Course Code	XMM_400445
Credits	6
Period	P1+2+3
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. S. Bhulai
Examiner	
Teaching Staff	
Teaching method(s)	

Course Objective

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Course Content

This course is part of the joint national master programme in mathematics. For schedules, course locations and course descriptions see https://elo.mastermath.nl.

Additional Information Teaching Methods

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Method of Assessment

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Additional Information Target Audience

mMath

Custom Course Registration

You have to register your participation in each Mastermath course via https://elo.mastermath.nl/login/. Registration is mandatory and absolutely necessary for transferring your grades from Mastermath to the administration of your university.

Dual Work Period

Course Code	XM_41010
Credits	12
Period	Ac. Year (September)
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	drs. H.J.M. van Goor-Balk
Examiner	prof. dr. S. Bhulai
Teaching Staff	dr. R. Bekker, dr. A. Zocca, dr. J. Berkhout, prof. dr. S. Bhulai, prof. dr. R.D. van der Mei, prof. dr. G.M. Koole
Teaching method(s)	Lecture

Course Objective

During the dual work period students gain experience and skills. The dual period allows students to bring learning into practice. In addition, students receive relevant work experience while studying and can already develop (soft) skills which are of great value after graduation. As a result, student are easier to deploy in the labor

market after graduation. At the end of the dual work period students:

- have gained experience and skills regarding a Business Analytics profession;
- have been able to apply their theoretical knowledge into practice;
- have brought practical work experience back to the university;

- have developed (soft) skills.

In particular, students furthermore:

- show interest in dual workplace;
- are committed to the organization/department;
- take initiatives for the dual workplace;
- make agreements and respects those;
- are aware of their duties, rights and obligations;
- take their responsibilities;
- can communicate appropriately;
- are open to questions and suggestions;
- are able to cooperate;
- have professional knowledge;
- make positive contribution to cooperation processes;
- are capable of problem solving;
- are able to develop;
- use reflection and evaluation;
- are capable to organize and plan their tasks, and to implement;
- are capable to perform under time pressure.

Course Content

Over a period of sixteen months, students are required to divide their time equally between work and study. Students start the dual work period from 1 September and completes it 16 months later on 31 December. Study and work are fully integrated. Students are employees and students at the same time. Students are on the payroll of the host organization for 20 (at most 24) hours a week during these 16 months. Students will conduct work which is of direct relevance to the master program Business Analytics.

Additional Information Teaching Methods

During the dual masters Business Analytics:

- students are employed by the host company on a fixed-term contract and is therefore officially on that company's payroll;
- student's working activities will be closely involved with the host organization's primary business processes;
- students will often be required to work with others in a team setting, whereby it may not be possible to identify clearly the student’s own contribution or the products for which students are (individually) responsible;
- a final report written by the student thus serves an important role in assessing the student’s clear understanding of the teamwork and products;
- the student's work period is of fixed duration: 20 hours a week (at most 24 hours a week) for 16 months.

Method of Assessment

Students need to submit a digital version of the interim report before March 31. Students will receive feedback on this interim report in a face-to-face meeting with one of the VU departmental staff members of the Mathematics or Computer Science department, so students are able to adjust their activities to a higher level in consultation with the external supervisor.

Furthermore, students are required to produce a written report of their activities during the entire work period, and to give a presentation at the Vrije Universiteit Amsterdam at the end of their dual work period (December or January). The report and presentation should clearly demonstrate an understanding of the structure of the organization, and will describe the activities students have undertaken and the project(s) in which students took part, as well as the relationship between these activities and the Business Analytics program. If students are involved in more than one project, each should be described separately briefly, and importantly, one project should be described in depth demonstrating the student’s mastery of the Business Analytics technique(s) that students used, as well as the student’s ability to communicate this knowledge to others.

A staff member of the Faculty of Science will, in consultation with the external supervisor, give an overall assessment of the work period. This assessment will take into account:

- the level of the required activities: based on the final report and the presentation (55% of the final grade);
- the level and the manner in which the required activities were performed: the assessment forms to be completed by the external supervisor and the meeting halfway through the dual work period (45% of the final grade).

Both components of the assessment need to be minimally satisfactory: one component cannot compensate the other.

There is no re-sit opportunity for the course "Dual Work Period".

Additional Information Target Audience

Participation is restricted to students Dual track of the master programme Business Analytics

Additional Information

More information is available on the website of theInternship Office Business Analytics.

Custom Course Registration

Admission to the dual master program is granted to those who have a Business Analytics Bachelor degree. For those with another university Bachelor degree, such as Mathematics, Artificial Intelligence, Econometrics, Computer Science, or a Bachelor degree from a university of applied sciences in the Netherlands (HBO), admission may be granted on an individual basis.

The admission requirements for the dual master programme in Business Analytics are almost identical to the admission requirements for the master programme in Business Analytics. The only difference is obtaining suitable employment. Those seeking admission to the dual master program should realize that admission also depends on obtaining suitable employment. The Vrije Universiteit Amsterdam has contacts with a number of companies that are interested in participating in this program. Be aware you cannot work full-time during your dual master Business Analytics. Students are on the payroll of the host organization for 20 (at most 24) hours a week during the Dual Work Period. The formal approval of one of the Business Analytics staff members is required before students can actually take up employment.

Explanation Canvas

A registration for this course automatically gives access to the corresponding Canvas website.

Dynamic Programming and Reinforcement Learning

Course Code	XM_0093
Credits	6
Period	P2
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. G.M. Koole
Examiner	prof. dr. G.M. Koole
Teaching Staff	V.G.M. Francois Lavet, prof. dr. G.M. Koole
Teaching method(s)	Seminar, Lecture

Course Objective

Dynamic Programming (DP) and Reinforcement Learning (RL) are fields concerned with decision making over time. After completing this course, the student

- is familiar with the commonly used algorithms for solving dynamic optimization problems;
- understands the main features of these algorithms, their strengths and weaknesses including their convergence properties;
- can implement them in an appropriate language;
- can model real-world decision problems into a DP or RL framework and solve moderately sized problems;
- has knowledge of the historical development of DP and RL and has an idea of possible future developments.

Course Content

This course is concerned with reinforcement learning and its origin dynamic programming. These are fields dealing with goal-directed decision making over time, such as finding your way in an unknown area, playing a game or pricing airline tickets.
We look at these areas from different angles:

- we deal with full-information "planning" problems, but also with partial-information "learning" problems
- we consider different algorithms, some of which are guaranteed to find the best solution, but also heuristics
- we consider high-dimensional problems (such as games) and methods to solve them

- we look at small toy problems to understand algorithms and sharpen our intuition, but also bigger problems for which we learn how to implement algorithms (in python)
- we look at different types of applications, both from AI (search problems, games) and OR

Additional Information Teaching Methods

Lectures and practical work integrated

Method of Assessment

Programming exercises and final exam.
The 3 assignments each count for 10% and the exam for 70%.
The minimal passing grade for the exam is 5.0.

Literature

Slides and lecture notes

Additional Information Target Audience

mBA, mAI, mCS, mBa-D, mMath

Recommended background knowledge

Programming experience in python

Entrepreneurship in Analytics and AI

Course Code	XM_0090
Credits	6
Period	P4
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. W. Stam
Examiner	prof. dr. W. Stam
Teaching Staff	prof. dr. W. Stam, H.V. Nguyen, drs. G.J. Beekman MA, P. Afonso Branco Ramos Pinto
Teaching method(s)	Seminar, Written partial exam, Lecture

Course Objective

Academic and research skills: Students are able to generate a promising new business idea in the field of AI & Analytics and develop a prototype that validates the proposed customer needs and market potential.

Bridging theory and practice: Students understand the core theories and concepts in the field of entrepreneurship, as reflected by their ability to analyze, compare, and apply these theories and concepts to analyze and solve the key management challenges faced by real-world entrepreneurs.

Entrepreneurial soft skills: Students are able to effectively work in a team for generating and developing a new venture idea and can pitch the business concept to external stakeholders in a professional, engaging manner.

Course Content

Entrepreneurship in Analytics and AI is an engaging course for students who like to learn about entrepreneurship, its role and importance in our society, and the process by which entrepreneurs transform new ideas into successful AI-driven business ventures.

In this course, students will learn the ropes of this emerging field via three educational pillars:

1. Students will learn to understand and apply theories from economics, sociology, and psychology to study key topics in entrepreneurship, including innovation, new venture creation, entrepreneurial competences, entrepreneurial finance, scaling and growth of new ventures, corporate entrepreneurship, and entrepreneurial ecosystems. Theoretical understanding of these subjects is applied to real world cases focusing on key management challenges faced by entrepreneurs and how to creatively overcome these challenges.

2. Students will familiarize with a set of entrepreneurial soft skills, which they will put into practice when they will be asked to approach different stakeholders to further their team projects to gain commitments or to obtain important information. For this purpose, the students will receive training in generating initial ideas for opportunities as well as networking and network utilization.
3. Students will be introduced to the actions and strategies that promote the process of new venture creation. For this purpose, the students, supported by the lecturers and coaching, will write (and pitch) a Student Design Sprint report (including prototype) around an innovative new venture opportunity in the field of Analytics and/or AI that they deem interesting for entrance into the market.

Additional Information Teaching Methods

During this course, students will attend both (guest) lectures and tutorials. The lectures explain relevant concepts, theories, and frameworks in the field of entrepreneurship and illustrate their usefulness using various practical examples. The tutorials are designed to facilitate students in making progress with their team projects by providing appropriate guidance and feedback.

Method of Assessment

The final grade for this course is based on a written individual exam, several individual pass/fail assignments, and a Student Design Sprint (SDS) project assignment.

The written exam and pass/fail assignments are individual assessments whereas the SDS assignment is a team assessment. To pass the course, the exam and SDS assignment should be graded as sufficient. Both forms of assessment have one option for re-sit in case students received an insufficient score.

Literature

The literature for this course constitutes a selection of academic articles and case studies.

Evolutionary Computing

Course Code	X_400111
Credits	6
Period	P1
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. A.E. Eiben
Examiner	prof. dr. A.E. Eiben
Teaching Staff	prof. dr. A.E. Eiben, mr. J.M. di Matteo, dr. K. da Silva Miras de Araujo
Teaching method(s)	Seminar, Lecture

Course Objective

This course has a threefold objective:

1. (Knowledge & Understanding) To learn about computational methods based on Darwinian principles of evolution.
2. (Applying knowledge and understanding, Communication) To use such methods as problem solvers and as simulation tools; to determine adequate algorithm setups depending on the problem at hand.
3. (Applying knowledge and understanding, Lifelong learning skills) To gain hands-on experience in performing computational experiments with evolutionary algorithms.

Course Content

This course is about constructing, applying and studying algorithms based on the Darwinian evolution theory. Driven by selection (survival of the fittest, mating of the fittest) and randomised reproduction (mutation, recombination), an evolutionary process is being emulated and solutions for a given problem are being "bred". During this course, various flavours within evolutionary computing are treated, including genetic algorithms, evolution strategies, evolutionary programming, genetic programming, differential evolution, particle swarm optimisation. Applications in optimisation, constraint handling, machine learning, and robotics are discussed. Specific subjects handled include: genetic structures (representations), selection techniques, sexual and asexual reproduction operators, (self-)adaptivity and methodological aspects, such as algorithm design & tuning, and performance assessment. Special attention is paid to the field of evolutionary robotics where the bodies (morphology, hardware) and the brains (controller, software) of robots are evolved simultaneously. Hands-on-

experience with evolutionary algorithms is gained through a compulsory programming assignment.

Additional Information Teaching Methods

Oral lectures and compulsory Python programming assignment (in teams of 3 to 5). A limited number of highly motivated students can replace the programming assignment by a special research assignment under the personal supervision of the lecturer(s). These research projects aim at publications with the students as first authors of the paper.

Method of Assessment

Written exam and programming assignment (in teams of 3 to 5). To pass the course as a whole, you must pass both the exam and the programming assignment. For the final grade they will count with the same weight:

Grade = 0.5 x Exam_Grade + 0.5 x Assignment_Grade

Note that:

- The exam is multiple choice, taken on the campus in an electronic form. There is a resit in case you cannot attend or fail.
- The programming assignment is graded based on your code, the quality of the solution your algorithm finds, and a short report. **No resit is possible for the programming assignment.**
- If you pass one component (Exam or Assignment), but fail the other, then you can keep the grade of the passed component for next year or later. Then you can pass the course by getting a good grade (> 5.5) for the missing component and using the old grade in the averaging formula.

Literature

The course is based on the textbook ***A.E. Eiben and J.E. Smith, Introduction to Evolutionary Computing, Springer, 2015, 2nd edition, ISBN 978-3-662-44873-1***. The lectures and the book chapters are synchronised to a great extent. In addition, a number of carefully selected papers to provide extra information on specific topics are available via Canvas.

Additional Information Target Audience

MSc Artificial Intelligence
MSc Econometrics and Operations Research
MSc Finance
MSc Business Analytics
MSc Computer Science
MSc Computer Security
MSc Bioinformatics and Systems Biology

Additional Information

Attending the lectures in real life is not mandatory, but highly recommended. Active participation in the team work for the programming assignment is a must!

Explanation Canvas

The complete schedule of the course, organised by the chapters of the textbook, will be made available at the beginning of the course. Students are assumed to consult the course schedule to keep up with the progress, as Canvas rubrics are not used. Video recordings of the lectures made during the COVID period are still available on Canvas for self-study.

Recommended background knowledge

Python programming skills are necessary to do the practical assignment. If you cannot program, you cannot complete this course.

Financial Markets and Institutions

Course Code	E_FIN_FMI
Credits	6
Period	P2
Course Level	400

Language Of Tuition	English
Faculty	School of Business and Economics
Course Coordinator	A. Ivashchenko
Examiner	A. Ivashchenko
Teaching Staff	A. Ivashchenko, S. Rijks, S. Rijks
Teaching method(s)	Study Group, Lecture

Course Objective

This course aims to develop an understanding of the economics underlying financial markets, financial intermediation, and banking, with particular regard for current market developments and trends. By the end of this course, students should be able to describe formally the state of different financial markets and derive implications for the broader economy.

Course Content

The course is divided into five blocks that roughly correspond to study weeks:

1. Fixed Income Markets and Monetary Policy.
2. FX Markets and International Finance.
3. Banking, banking regulation, systemic risks.
4. Asset Management.
5. Commodity Markets.

We start with a quick refresher on the key elements of the modern financial system (cornerstone markets and institutions). The course goes on to focus, primarily, on financial markets beyond public equity. Thus, we take a closer look at fixed-income markets with an emphasis on interest-rate products and the role of monetary policy (weeks 1 and 2). Observing that interest rates are set differently in different currencies, we study modern FX markets as a link between interest rate markets in different currencies (week 3). We establish a crucial role that financial intermediaries play in financial markets and focus on the economics of banking and asset management to deepen our understanding of market mechanisms (weeks 4 and 5). We conclude the course by studying the financialization of commodities markets (week 6), which highlights yet again the interconnectedness of the financial and the real economy.

Additional Information Teaching Methods

Most study weeks in this course are structured as follows:

- Background readings and knowledge clips are uploaded on Canvas before the start of the week. The students are supposed to familiarize themselves with this material before we start digging deeper into the topic in the classroom.
- The lecture takes place early in the week. It aims to present a broad picture of the topic of interest and introduce key analytical concepts we practice later in the tutorial.
- The tutorial takes place toward the end of the week and is centered around a quantitative exercise that utilizes a practically helpful analytical concept. We aim here to both master the analytical part and to derive qualitative economic implications of our quantitative analysis.
- Every other week, there is a Q&A session and the end of the study week.

There are usually a few guest lectures in the class that change from year to year and aim to address topical financial markets issues. In past years, we had guest industry speakers talking about interest rate benchmarks reform, bank risk management, pension funds asset-liability management, mutual fund sales, and the practice of monetary policy.

We study the financialization of commodity markets in week six by solving a case study and running a (team) case competition.

Method of Assessment

The course grade is a weighted average of the case study solution (week 6, a group assignment) and the final exam. The weights are typically 0.35 and 0.65, respectively (but this is subject to change). The final exam mixes multiple-choice (circa 60% of the exam grade) and open questions (the remaining 40%). The exam is open-book. There are bonus points available throughout the course that add to the final grade.

Entry Requirements

Students should be familiar with the material of the bachelor course "Financial Markets and Institutions", or similar.

Literature

The lectures and tutorials of the course are generally self-contained and are the primary study materials for the class. Few compulsory readings complement the lecture material – they are listed below. The reading list might change as we proceed with the course; timely announcements about changes in the reading list will be made on Canvas.

Block 1, Fixed Income Markets and Monetary Policy:

- Greenwood, R. & Vissing-Jorgensen, A., 2018. The Impact of Pensions and Insurance on Global Yield Curves. Working Paper.
- Kim, D., Walsh, C. & Wei, M., 2019. "Tips from TIPS: Update and Discussions," FEDS Notes. Washington: Board of Governors of the Federal Reserve System

Block 2, FX Markets and International Finance:

- Borio, C., McCauley, R., McGuire, P. & Sushko, V., 2016. Covered interest parity lost: understanding the cross-currency basis. BIS Quarterly Review, Volume III, pp. 45-64.
- Du, W., Tepper, A. & Verdelhan, A., 2018. Deviations from Covered Interest Rate Parity. The Journal of Finance, 73(3), pp. 915-957.

Block 3, Banking, banking regulation, systemic risks:

- Diamond, D. W., 2007. Banks and Liquidity Creation: A Simple Exposition of the Diamond-Dybvig Model. Economic Quarterly, 93(2), pp. 89-200.
- Duffie, D., 2020. Still the World's Safe Haven? Redesigning the U.S. Treasury Market After the COVID-19 Crisis. Hutchins Center Working Paper, Issue 62.

Block 4, Asset Management:

- Lettau, M. & Madhavan, A., 2018. Exchange-Traded Funds 101 for Economists. Journal of Economic Perspectives, 32(1), pp. 135-154.
- Berk, J. B. & van Binsbergen, J. H., 2017. Mutual Funds in Equilibrium. Annual Review of Financial Economics, Volume 9, pp. 147-167 (see an ungated version on SSRN).

Block 5, Commodity Markets:

- Readings change each year and will be announced separately

Investments

Course Code	E_EBE3_INVES
Credits	6
Period	P5
Course Level	300
Language Of Tuition	English
Faculty	School of Business and Economics
Course Coordinator	prof. G.U. Weitzel
Examiner	prof. G.U. Weitzel
Teaching Staff	prof. G.U. Weitzel
Teaching method(s)	Seminar, Lecture

Course Objective

This course offers a comprehensive introduction to the world of investments. The key objective of this course is to provide understanding of the pricing of different asset classes and insights into the principles of investment analysis.

After successfully completing this course, you will be able to:

- understand and apply various approaches to price financial assets, such as equity and derivatives (Academic and Research Skills, Bridging Theory and Practice - Knowledge, and Bridging Theory and Practice - Application)
- understand and apply optimal asset allocations and fundamental risk-management techniques (Academic and Research Skills, Bridging Theory and Practice - Knowledge, and Bridging Theory and Practice - Application)

- understand the determinants and implications of market efficiency and financial decision making (Academic and Research Skills, Bridging Theory and Practice - Knowledge, and Bridging Theory and Practice - Application)
- acquire a critical attitude towards competing theories in investments (Academic and Research Skills, Bridging Theory and Practice - Knowledge, and Bridging Theory and Practice - Application)
- be able to do advanced analyses on data sets with real data in in R and RStudio (Academic and Research Skills, Bridging Theory and Practice - Knowledge, and Bridging Theory and Practice - Application)

Course Content

Investment decisions take a prominent role in everyday life and are regularly taken by individual households (additional pension savings, savings for children education, buying a house, etc.), and by institutional investors (banks and corporate firms, insurance companies, investment funds). By extending and deepening prior (undergraduate) Finance knowledge, this course, covers several areas of investments, including, Portfolio Theory and Asset Pricing, Portfolio Management, Risk Management, Security Valuation, and Derivatives. Throughout the course, students will not only learn theoretical concepts, but also apply the theory with real data in R. The course therefore builds on a combination of theory and empirics. Students will work on data assignments in which they can directly apply their newly acquired theoretical knowledge and experience the real-life issues associated with investing. The methods taught will also be helpful for analyses for bachelor and master theses.

Additional Information Teaching Methods

Lectures and tutorials.

Method of Assessment

- Written exam
- Online data assignments
- Online quizzes

Entry Requirements

Finance I or equivalent. Knowledge of Finance II or equivalent, of econometrics, and of R is highly recommended (also see below).

Literature

- Zvi Bodie, Alex Kane and Alan J. Marcus: Investments, McGraw Hill (Latest Global Edition).
- Additional readings and knowledge clips may be posted on Canvas.

Additional Information

This course provides theoretical and applied investments knowledge for students aiming at an MSc in Finance and a career in the financial sector.

Recommended background knowledge

The course relies on prior knowledge on linear algebra, statistics and econometrics (as covered in QRM I, QRM II, and QRM III). Even though the course offers a brief introduction to the concepts and tools in these fields, students are strongly advised to review this material from relevant courses in the first two years of studies. We expect students to have a good understanding of the material covered in Finance I and Finance II (and/or equivalent courses) and we recommend that students refresh/acquire basic R skills.

Learning Machines

Course Code	XM_0061
Credits	6
Period	P3, P6
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. K. da Silva Miras de Araujo
Examiner	dr. K. da Silva Miras de Araujo
Teaching Staff	dr. K.S. Luck, dr. K. da Silva Miras de Araujo, dr. A. Yaman
Teaching method(s)	Seminar, Lecture

Course Objective

1. Understanding and being able to critically discuss how learning machines relates to machine learning: This includes the main aspects involved in both, as well as their implications. (Knowledge and Understanding)
2. Designing systems of robots able to adapt to solve tasks: This includes endowing a robot with learning mechanisms that make use of sensing capabilities. (Applying Knowledge and Understanding)
3. Designing and conducting experiments with simulated and real robots to solve tasks. (Applying Knowledge and Understanding)
4. Being able to analyze the results from experiments (3), deriving insights into diverse aspects related to the methodology utilized. (Judgement)

Course Content

- This course concerns **designing robot systems** that can sense their environment, act upon it, and improve their behavior to solve tasks.
- **Most** of the course encompasses **practical sessions** with only a few lectures.
- Multiple tasks should be delivered, and students have the freedom to choose different learning and optimization methods to work with.
- All tasks will be done in **simulation AND in hardware** using Robobo robots. The Robobos are wheeled robots with fixed bodies that possess infrared sensors and a camera.
- Students are expected to **design/optimize the controllers (brains)** of the robots to solve the due tasks while utilizing the sensors as inputs.
- For the simulations, a robot programming/simulation framework will be provided to students. The **simulator** utilized is VREP (Coppelia).

Additional Information Teaching Methods

- The course is short and requires **intensive work** - it happens over a period of **four weeks** - each week contains one **lecture**, two **practical** sessions, and one **presentation** & evaluation session.
- **Attendance at all sessions is mandatory.** Missing lectures and practical sessions must be justified to the teacher and might be very detrimental to the assignment of the week, while missing a presentation incurs failing the assignment of that week.
- Students will work in **teams**.
- This is a course for students willing to go the extra mile, requiring a high level of **independence**.

Method of Assessment

1. Task I (presentation and demo): 20/100
2. Task II (presentation and demo): 20/100
3. Task III (presentation and demo): 20/100
4. Final report: 40/100

Tasks scores range from 0 to 10, and **all tasks must be above 5.5 to pass the course**

In each presentation, the group is required to disclaim each member's contribution explicitly. Additionally, each member must participate in presenting and, most importantly, in answering questions from the teacher or TA during/after the presentation. If the student does not **demonstrate enough individual knowledge**, their score for that assignment will be individually penalized.

The course does **not offer resits**, but students can **reuse assignment's grades** in case they fail and enrol again.

Additional Information Target Audience

Master Artificial Intelligence
Master Business Analytics

Recommended background knowledge

- Having passed any course involving **learning algorithms**, e.g., Reinforcement Learning, Evolutionary Computing, Computational Intelligence, Deep Learning, etc. - the course will not teach you new algorithms and it is fundamental that you have intermediate skills with at least one learning algorithm.
- Intermediate **programming** skills
- Python
- Independence for self-study

Not mandatory, but useful: Experience with robotics simulators - these can be very challenging to work with.

Machine Learning for the Quantified Self

Course Code	XM_40012
Credits	6
Period	P6
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. M. Hoogendoorn
Examiner	prof. dr. M. Hoogendoorn
Teaching Staff	V.G.M. Francois Lavet, prof. dr. M. Hoogendoorn, B. Borsboom
Teaching method(s)	Practical, Lecture

Course Objective

The main aims of this course are to make the student familiar with specific machine learning techniques for quantified self and sensory data. Both the theory as well as the practical side of matters will be addressed. More specifically, the following aims are distinguished:

1. Understanding the domain of the quantified self and sensory data
2. Knowing and being able to apply techniques for outlier detection on sensory data.
3. Knowing and being able to apply techniques for feature engineering on sensory data.
4. Knowing and being able to apply clustering techniques related to sensory data.
5. Understanding the theory of supervised learning and its implications.
6. Learning how to apply non-temporal machine learning techniques to sensory data.
7. Knowing and being able to apply temporal machine learning techniques to sensory data, including deep learning techniques.
8. Knowing and being able to apply reinforcement learning in the application of the quantified self.

The course if focused on the Dublin descriptors Knowledge and understanding (learning new relevant algorithm in this domain); Applying Knowledge and Understanding (since the aim is also to apply the algorithms in practice); Making judgements (which technique is appropriate, how to best apply the techniques for a specific case); Communication skills (how to report on your approach, choices and your results), and Learning skills (able to find new relevant techniques, assess their suitability, etc.).

Course Content

The quantified-self refers to large-scale data collection of a user's behavior and context via a range of sensory devices, including smart phones, smart watches, ambient sensors, etc. These measurements contain a wealth of information that can be extracted by means of machine learning techniques, for instance for the purpose of predictive modeling. In addition, machine learning techniques can be a driver for adaptive systems to support users in a personalized way based on the aforementioned measurements. The type of data does however require specialized machine learning techniques to fully exploit the information contained in the data. Examples of challenges include the temporal nature of the data, the variety in the type of data, the different granularity of various sensors, noise, etcetera.

In this course specific techniques to handle quantified self (or broader sensory data) will be treated. More in specific, it will address:

- Feature engineering (how do we come from raw data to usable features):
  * Removing noise from data
  * Handling missing data
  * Identifying (temporal) features
- Learning of user patterns:
  * Temporal machine learning approaches such as recurrent neural networks, time series analysis
  * Clustering approaches with dedicated distance metrics (including dynamic time warping)
- Adaptive feedback and support
  * Reinforcement learning
- Integration of the various components.

In addition, a number of real-life applications will be discussed. Next to lectures, there will be an extensive practical part, where students will learn to work with various algorithms work on a project they propose themselves.

Additional Information Teaching Methods

The course will be taught in four weeks. Both regular lectures (l, first two weeks) and practical sessions (p) will be organized. During the first two weeks the emphasis will be on the lectures and getting started with the practical assignment that ties closely to the content of the lectures. The third week will be working on the practical assignment fulltime. In the fourth week a mandatory final lecture will take place in which a selection of groups will

present their project. The course ends with an exam.

Method of Assessment

Written exam (50%) and practical assignment (50%). For both parts, the grade needs to be sufficient to obtain a final grade. The practical assignment is done in groups of three. For the practical assignment, two intermediate versions of the report should be submitted for feedback that are given a pass/fail. Only in case of a pass for the two intermediate versions will a final grade be given for the practical assignment.

The written exam will be multiple choice and have a resit during the Summer period. On top, in case your assignment grade average is insufficient you are allowed to redo the assignments by improving it based on the feedback you received. The maximum grade for this resubmitted assignment is set to a 6.

Literature

Hoogendoorn, M. and Funk, B. Machine Learning for the Quantified Self - On the Art of Learning from Sensory Data, ISBN 978-3-319-66308-1, Springer Verlag 2018. The book can be downloaded free of charge through the UBVU, just follow the link to the book: http://dx.doi.org/10.1007/978-3-319-66308-1

Papers on state-of-the-art deep learning techniques.

Additional Information Target Audience

Master Artificial Intelligence, Master Business Analytics, Master Computer Science

Additional Information

Lecturer: Prof.dr. M. Hoogendoorn

Custom Course Registration

Registration for medical informatics (UvA) students: Please check the following document https://www.dropbox.com/s/a2kd7xacuxt5y8e/AI%20for%20Health%202022-2023.
and then register for secondary courses via https://vu.nl/en/student/information-about-registration-and-enrolment/in

Recommended background knowledge

Programming experience. Knowledge of basic machine learning algorithms.

Master Project Business Analytics

Course Code	X_400459
Credits	36
Period	Ac. Year (September)
Course Level	600
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	drs. H.J.M. van Goor-Balk
Examiner	prof. dr. S. Bhulai
Teaching Staff	drs. H.J.M. van Goor-Balk
Teaching method(s)	Lecture

Course Objective

The Master Project Business Analytics is the graduation project for the Master Business Analytics or the Dual Master Business Analytics. As the graduation project this course builds on and integrates previous courses on Business Analytics. Students show that they are capable of independent research at an academic level (typically applied in a practical context) on a specific topic in the field of Business Analytics under supervision of one of the staff members of the Faculty of Science. Furthermore, students write an academic report to present the applied scientific methods and the obtained results. Also, students give an oral presentation that is tailored to the audience. As such, the skills described below should be demonstrated during the Master Project Business Analytics.

Academic (research) skills:

- Students formulate a research question.

- Students study relevant literature.
- Students either develop or select algorithms to generate an accurate model based on data related to the problem or develops a (mathematical) model of the problem.
- Students analyze the model using skills and techniques acquired during the master's program Business Analytics, and shows that they are able to connect theories, models, and concepts with and within the research topic.
- Students answer the research question using analytical and numerical methods.
- Students are capable of independent and critical reflection on their own and on other relevant work.
- Students are able to write a report that clearly describes the research question, the motivation for this research, the relevant literature, the used methodology, the results, conclusions and recommendations.
- Students are able to give a clear presentation about the research question, the motivation for this research, the used methodology, the results, conclusions and recommendations.
- Students are able to collect data for validating and / or fitting the model, and implement and verify computer programs.

Field of work:

- Students take responsibility for their own actions.
- Students show that they are capable of making well-founded decisions and can support these choices.
- Students show an appropriate work attitude during the master project, e.g. independence, initiative, enthusiasm, punctuality.
- Students produce the required products and/or reports according to the schedule.
- Students gain an understanding of the way in which the organization functions.
- Students practice and display social and communication skills.
- Students explore potential career options.
- Students maintain and strengthens contacts between the Vrije Universiteit Amsterdam and the host organization.

Course Content

The Master program in Business Analytics is concluded with the external Master Project Business Analytics. This project differs from all other parts of the study program as students carry out a full-time research project for six months at an external organization. The project must scientifically be state of the art and must also have clear educational value. Students will work on a predefined project and are required to create the required products within the stated period. The internship proposal must be approved in advance by the Internship Board and first supervisor.

Additional Information Teaching Methods

Students are an intern of the host organization for at least six months on a full-time basis. Students will be supervised by one staff member of the Faculty of Science and a supervisor from the host company. A second reviewer of the Faculty of Science will need to approve the plan of action, and will assess the final thesis. Students also take part in the Back to the VU Event every other month (in total 3 x 3,5 hours lecture). During the final session, students will give a short presentation about their own Master Project Business Analytics research. Attendance is mandatory.

Method of Assessment

The assessment method consists of a written report (55%), an oral presentation at the host company (10%), and the execution and attitude (35%). The internship report, which is also the student's formal thesis, is a written report about all relevant aspects, including the final product and all relevant scientific background and motivation. The internship will be concluded with a formal oral presentation, usually at the host organization's premises. Students have to submit the written report and give the presentation no later than the finishing date of the internship. After consultation with all supervisors involved, students may submit the report and/or give the presentation within two months after completion of the internship. After those two months students must have an adequate grade (6 to 10), otherwise the internship will be counted as not accomplished.

The thesis and final presentation are the only parts of the assessment that can be improved when initially graded as insufficient.

Entry Requirements

Before starting an internship and/or graduation project, students must have earned the credits of XM_0054 Research Seminar Business Analytics and all other programme credits except a shortfall of 6EC is permissible. If students wish to start on the internship and/or graduation project in January (2026), an additional shortfall of 6EC is permissible provided it is the credits of a course students have followed in period 2 or period 1 and 2 (of 2025-2026), have taken part in the examination and are expecting the result to be announced in January (2026).

To have the above requirement checked, students must contact the master coordinator before the start of the

graduation internship with the list of marks attached.

Students are required to read the Course Manual which is available on the Master Project Business Analytics Canvas website and to take the required steps mentioned in the Course Manual.

Literature

Assigned individually.

Additional Information Target Audience

mBA, mBA-D

Custom Course Registration

Students have to register themselves for the Master Project Business Analytics via the registration module on vu.nl.

Explanation Canvas

If students are planning to start their Master Project Business Analytics within four months and they cannot register for the Master Project because they do not meet the requirements yet, please contact Annemieke van Goor (H.J.M.van.Goor-Balk@vu.nl) so she can give access to the Master Project Business Analytics Canvas website. Here students can find all the information they need about the Master Project Business Analytics.

Mathematical Optimization

Course Code	XM_0051
Credits	6
Period	P2
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. R. Paradiso
Examiner	dr. R. Paradiso
Teaching Staff	dr. R. Paradiso, dr. A. Zocca
Teaching method(s)	Seminar, Computer lab, Lecture

Course Objective

After completing this course:

- You can formulate a problem as an optimization model in a correct and efficient way.
- You understand when you need to use discrete decision variables and the impact of those in the effort required to solve problems.
- You can prove that a given model is convex, and you can characterize the optimal solutions and formulate the equivalent dual model.
- You can reformulate most of the convex models into Second Order Conic Optimization models that can be efficiently solved.
- You are able to formulate and solve optimization problems that do account for uncertainty, both with a stochastic and a robust approach. You should be able to translate (when possible) robust or stochastic requirements into equivalent but finite constraints that you can solve.
- You can solve small problems of all the types studied on paper and problems of practical size on the computer using the Python programming language.

Course Content

Mathematical optimization is among the most essential instruments of prescriptive analytics and is used to make optimal decisions based on quantitative arguments.

This course will make you familiar with translating practical problems into optimization models and solving those models. Despite the goal of modelling and solving such problems, this course teaches the fundamental results from the mathematics of optimization, including optimality conditions, duality, stochastic optimization, and robust optimization.

The course covers linear optimization and its generalizations (conic and convex optimization). We will also consider

optimization under uncertainty. Optimization models will be implemented and solved in Python using both open-source solvers and commercial solvers that are free for education.

Additional Information Teaching Methods

The course consists of the following contact moments:

Lectures in 2 blocks of 1:45 per week.
Tutorials in 1 block of 1:45 per week.
Computer lab in 1 block of 1:45 per week.

Presence and active participation is expected during all the above.

Method of Assessment

Exam (60%)
Practical assignments (40%)
The practical assignments cannot be repeated. To pass:

- The grade for the exam needs to be 5.0 or higher,
- The weighted average needs to be 5.5 or higher.

Literature

Lecture notes, slides, and additional material published via Canvas

Additional Information Target Audience

Students of Business Analytics, Econometrics and Operations Research, or other master programs with a strong quantitative accent interested in Mathematical Optimization theory and practice.

This course is not intended for students who followed the course Mathematical Optimization in their minor due to considerable overlap.

Explanation Canvas

The slides, assignments and additional information will be made available via Canvas.

Recommended background knowledge

A first course in Operations Research and Linear Algebra is required background.

Besides introduction to Operations Research and Linear Algebra you should be familiar with (basic) programming in Python.

Multi-Agent Systems

Course Code	XM_0052
Credits	6
Period	P2
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. E.J.E. Pauwels
Examiner	dr. E.J.E. Pauwels
Teaching Staff	prof. dr. K.V. Hindriks, dr. E.J.E. Pauwels
Teaching method(s)	Seminar, Lecture

Course Objective

After successfully completing this course, the student

- Has a solid understanding of concepts from elementary and intermediate game theory, such as Nash equilibria for simultaneous games, backward induction and subgame perfect equilibria for sequential games and the Shapley value in cooperative games.

- Understands various principled approaches to balance exploration and exploitation;
- Has a solid understanding of the tabular solution methods for (single agent) reinforcement learning;
- Is able to explore and digest current research on deep reinforcement learning and multi-agent reinforcement learning.

Dublin descriptors:
1 Knowledge and Understanding,
2. Applying Knowledge and Understanding:

Course Content

In Multi-agent systems (MAS) one studies collections of interacting, strategic and intelligent agents.

These agents typically can sense both other agents and their environment, reason about what they perceive, and plan and carry out actions to achieve specific goals. In this course we introduce a number of fundamental scientific and engineering concepts that underpin the theoretical study of such multi-agent systems. In particular, we will cover the following topics:

- Agents: Typology and examples, strategic versus learning agents
- Introduction to non-cooperative game theory
- Introduction to coalitional game theory for teams of selfish agents
- Principles of Mechanism Design
- Exploration versus Exploitation
- Markov Decision Processes
- Reinforcement learning for a single agent
- Introduction to multi-agent reinforcement learning

Additional Information Teaching Methods

Two lectures (1h45) and one recitation class (1h45) per week.

Method of Assessment

There will be weekly homework assignments that will be graded (5 of which are done in groups and 1 is done individually). In addition, there will be a final exam that will test the student's ability to apply the course material to new and concrete problems.

The final grade will be a weighted average of the grades for the homework assignments (50%) and the final exam (50%).

There is a resit offered for the final exam and the individual homework assignment. There is no resit for the group homework assignments.

Literature

Recommended reading:

Yoav Shoham, Kevin Leyton-Brown: Multiagent Systems
Publisher: Cambridge University Press (15 Dec. 2008)
ISBN-10: 0521899435
ISBN-13: 978-0521899437

R.S. Sutton, A.G. Barto, F. Bach: Reinforcement Learning: An Introduction
Publisher: MIT Press; second edition edition (23 Nov. 2018)
Language: English
ISBN-10: 0262039249
ISBN-13: 978-0262039246

Additional Information Target Audience

Master Artificial Intelligence

Master Business Analytics

Recommended background knowledge

Basic calculus, probability theory and linear algebra. Fluency in a programming language.

Natural Language Processing

Course Code	XM_0121
Credits	6
Period	P5
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. L.E. Donatelli
Examiner	dr. L.E. Donatelli
Teaching Staff	dr. L.E. Donatelli
Teaching method(s)	Seminar, Lecture

Course Objective

Natural language processing (NLP) is a highly dynamic research field that aims at analyzing natural language using computational models and methods. This course provides an introduction to the principles of the field and has two main objectives:

1) Students learn the fundamentals for developing a system for a common NLP task.

- They can analyze and interpret the linguistic information provided by common pre-processing pipelines and are familiar with the core NLP terminology. (Knowledge & Understanding)
- They can describe a common NLP classification setup and evaluate the results. (Making Judgements)
- They can explain the functionality of neural networks and their use in NLP. (Communication)
- They can explain the role of pre-trained language models in NLP and apply them to a task. (Applying knowledge and understanding)

2) Students learn to systematically analyze and interpret NLP models for a specific task.

- They can describe the linguistic challenges associated with the task. (Making judgments)
- They can describe how the task is approached by a state-of-the-art model. (Knowledge and understanding)
- They are able to understand, evaluate, and interpret the output of a model. (Applying knowledge and understanding)
- They can compare the output of different models and analyze their strengths and weaknesses. (Lifelong learning skills)

Course Content

Natural Language Processing operates on the interface between linguistics and computer science. In order to get computers to deal well with natural language, it is important to understand both how language works and how computational methods work. Computational linguists work on this interface and have developed methods and technologies for language analysis.

This course covers technologies and computational models for core domains of natural language processing. Students are trained to find, process, and understand the latest developments in this rapidly advancing field. The course includes practical components that require students to work with Python code to explore NLP models. They learn to apply them to new data and reflect on the effect of modeling decisions on the performance.

Additional Information Teaching Methods

The course consists of two sessions per week. We generally alternate between more theoretical lectures and interactive practical classes in smaller groups.

Method of Assessment

The course is evaluated by practical assignments (50%) and an exam (50%). Both components need to be graded at least with a 5.5. Retakes of practical assignments to improve the grade are not allowed. Practical assignments might be organized as a mix of group projects and individual contributions.

Literature

This information will be specified on Canvas.

Additional Information Target Audience

Master Artificial Intelligence, also suitable for Master Business Analytics and other Master students in Computer Science

Recommended background knowledge

Students must have acquired programming skills in Python. For students with no background knowledge of machine learning, additional reading might be required as the fundamental concepts will only be briefly introduced in the course.

Optimization of Business Processes

Course Code	X_400422
Credits	6
Period	P4+5
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. G.M. Koole
Examiner	prof. dr. G.M. Koole
Teaching Staff	prof. dr. G.M. Koole
Teaching method(s)	Lecture

Course Objective

In this course students learn about applications of operations research in manufacturing and services, especially health care, call centers and revenue management.

After this course the students:

- are familiar with some areas in which business process optimization techniques are used;
- are familiar with the most common optimization techniques and tools and knows how to apply them;
- understand the business context and is familiar with the most common pitfalls.

Course Content

We deal with a number of important application areas of optimization:
production logistics/manufacturing, call centers, health care and revenue management.
For each area we present quantitative problems and discuss how they can be solved using mathematical models, mostly forecasting, deterministic optimization and stochastic processes. We also discuss a number of new models. Students solve representative problems from the field in homework assignments, which include all steps from data to decision. Several guest lectures are given by people from industry.

Additional Information Teaching Methods

Lectures and practical work.

Method of Assessment

Written examination, 3 individual assignments in R or python, and a book presentation.

The assignments and the presentation each count for 10%, the exam for 60%. The minimal grade for the exam is 5.0.

Literature

Lecture notes.

Additional Information Target Audience

mBA, mBA-D, mMath

Recommended background knowledge

Applied Stochastic Modeling or equivalent knowledge

Performance of Networked Systems

Course Code	X_405105
Credits	6
Period	P2
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. R.D. van der Mei
Examiner	prof. dr. R.D. van der Mei
Teaching Staff	prof. dr. R.D. van der Mei
Teaching method(s)	Lecture

Course Objective

Students will acquire basic knowledge of:

- Performance issues in communication systems, a complex interplay between communication networks (TCP/IP, UDP), application servers (HTTP, QUIC), services and clients
- Performance models for wireless and mobile network protocols (WiFi, GSM, GPRS, UMTS, HSDPA, 4G, 5G, 6G and mobile ad-hoc networks)
- Performance engineering principles and management of traffic in IP-based networks
- Performance modeling of streaming and elastic traffic
- Quantitative models for planning capacity in communication networks

Students will gain experience in modelling, engineering and planning performance of networked systems, and will learn how to tackle performance problems arising in today's information and communication systems.

Course Content

Over the past few decades, information and communication technology (ICT) has become ubiquitous and globally interconnected. As a consequence, our ICT systems are expected to process huge amounts of digital information, which puts a tremendous burden on our ICT infrastructure. At the same time, our modern society has become largely dependent on the well-functioning of our ICT systems: large-scale system failures and perceivable Quality of Service (QoS) degradation may completely disrupt our daily lives and have huge impact on our economy. These developments raise the need for understanding the nature and source of performance problems, and for finding effective ways to make our networks robust and reliable.

Motivated by this, the course will address performance-related challenges and solution methods in communication networks. To this end, we focus on performance modeling, traffic management, QoS provisioning and capacity planning for both wired and wireless/mobile communication networks (such as WiFi, 2G, 3G, 4G, 5G, 6G networks, and mobile ad-hoc networks). We address questions like:

- How do wired and wireless communication network technologies work?
- How can we model their performance?
- How can we manage and control traffic flows and provision QoS in IP-based networks?
- How can we deal with the omnipresence of uncertainty in virtually all aspects of communication networks?
- What techniques are available to model, analyze and optimize performance?

Additional Information Teaching Methods

Classroom lectures and a practical take-home homework assignment.

Method of Assessment

The assessment will be based on the homework assignment (total 20%) and a written exam (80%). Homework assignments can be made in groups of one or two students.

Literature

- A textbook on "High Performance Browser Networking" by Ilya Grigorik, O-Reilly, 2013.
- A reader on Stochastic Performance Modelling.
- Individual papers on specific topics.

Additional Information Target Audience

mBA, mCS, mPDCS, mCLS, mEctrie

Explanation Canvas

The relevant course information will be shared via CANVAS.

Recommended background knowledge

The students should have basic knowledge of computer networks.

Programming Large-Scale Parallel Systems

Course Code	XM_40017
Credits	6
Period	P1
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. F. Verdugo Rojano
Examiner	dr. F. Verdugo Rojano
Teaching Staff	dr. F. Verdugo Rojano, mr. M. Cassar
Teaching method(s)	Lecture

Course Objective

At the end of the course, students will be able to

- Understand the main principles of High Performance Computing (Knowledge and understanding)
- Design and implement efficient parallel algorithms on distributed-memory parallel computers (Applying knowledge and understanding)
- Make use of different distributed parallel programming models to implement these algorithms, including message passing with MPI and the task-based model based on remote procedure calls available in the Julia programming language (Applying knowledge and understanding).
- Analyze the performance of parallel applications and network topologies (Making judgements)
- Identify the main parallelization challenges in selected scientific computing applications (Knowledge and understanding)

Course Content

In this course, students learn how to design and implement computer programs that solve a problem in parallel on a distributed-memory computing system such as the supercomputers available in the top500 list (https://top500.org/). This is key in many real world applications to reduce execution time and to solve very large problems that cannot fit in a single compute node. You will learn how to design such parallel algorithms in order to minimize different types of performance overheads such as communication overhead, search overhead, and load imbalance. You will also learn how to analyze the theoretical and experimental performance of the resulting codes and the underlying computer network topology. Among all parallel programming models available for high performance computing, this course focuses on distributed-memory parallelization. We consider different distributed parallel programming models, including established techniques such as message-passing with MPI, but also modern approaches like the task-based model available in the Julia programming language. The course also discusses the application of these techniques to a set of real-world scientific computing applications, including partial-differential equations in solid and fluid dynamics, and n-body problems in astrophysics.

Additional Information Teaching Methods

- There are 2 class sessions per week. Part of the course follows the flipped classroom method. Pre-recordings, lecture notes, and exercises are made available in advance for self study, and follow up discussions take place in class meetings to deepen understanding.
- The course also includes the opportunity to run experiments on the DAS-5 computer cluster (http://www.cs.vu.nl/das5).
- Computer codes used in the course are implemented in the Julia programming language. Previous knowledge of Julia is not required as it will be learned as part of the course.

Method of Assessment

There is a final written exam (E) and a programming assignment (A). Their grades are combined as G = 0.1*A + W(E)*E, where

- W(E) = 1 for E < 7, and
- W(E) = 1-(E-7)/30 otherwise.

The grades E and A are at most 10 points each.

According to this formula the programming assignment is "quasi optional". There is little or no penalty for skipping the assignment in most of the cases. However, high-scoring students in the exam need the assignment for a perfect score.

Literature

Course material will be made available on Canvas. Part of the course is based on the lecture notes available at https://www.francescverdugo.com/XM_40017/dev/.

Additional Information Target Audience

Master Business Analytics
Master Computational Science
Master Computer Science
Master Computer Security

Additional Information

There is also a separate course "Parallel Programming Practical" (6 EC) in Period 3, the aim of which is to complement the contents of this course with further practical skills and experience.

Project Optimization of Business Processes

Course Code	X_400213
Credits	6
Period	P3
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. G.M. Koole
Examiner	prof. dr. G.M. Koole
Teaching Staff	prof. dr. G.M. Koole
Teaching method(s)	Practical, Lecture

Course Objective

Acquiring skills and experience necessary for building dashboards and decision support systems, and learning to apply relevant scientific knowledge.
Specifically, during the course you will:

- select the appropriate scientific method(s) for the problem at hand
- implement and validate the scientific methods chosen
- build a user-friendly interface for potential users in the field

The programming language will be decided upon with your supervisor.
Common programming languages used during the project are Python and R

Course Content

Project optimization of business processes concerns the construction and/or design of (part of) a dashboard or decision support system (DSS) that:

- is designed and built in a scientifically sound way;
- can be used in practice.

The dashboard/DSS is built in groups of students.

Additional Information Teaching Methods

Project

Method of Assessment

Grading for scientific quality, applicability, and verification of DSS, the project plan, and final report.
Individual grades for participation in the group are based on observed participation with the help of a time registration sheet and peer reviews.
There is no resit. If the DSS and report are of sufficient quality, but the individual contribution is insufficient by a small margin (max 1.0 pt), the student gets the opportunity to hand in an additional assignment to obtain a sufficient grade.

Literature

None.

Additional Information Target Audience

mBA, mBA-D

Additional Information

Important note: you are expected to attend the kick-off meeting. If (due to circumstances) you are not able to attend this meeting, you should notify the lecturer well in advance. Failing to do so may exclude you from this course. You are also expected to attend all progress meetings.

Recommended background knowledge

Applied Stochastic Modeling (X_400392).

Project Reinforcement Learning

Course Code	XM_0120
Credits	6
Period	P3
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	V.G.M. Francois Lavet
Examiner	V.G.M. Francois Lavet
Teaching Staff	V.G.M. Francois Lavet, prof. dr. M. Hoogendoorn
Teaching method(s)	Study Group, Lecture

Course Objective

Knowledge and understanding: students will be familiar with basic knowledge of (deep) reinforcement learning and they will be able to apply them to sequential decision-making tasks under uncertainty, including in the context of relatively high-dimensional inputs (e.g. time series).

Applying knowledge and understanding: students will be able to understand how to implement deep RL algorithms, how to compare them and how to apply them in real life problems.

Making judgements: students will have an understanding of the key learning techniques for sequential decision-making tasks used in RL, as well as an overview of a few key AI applications for such techniques along with a short discussion on societal impacts.

Communication skills: students will be able to write a scientific report about their project (in a group of 2-3 students).

Learning skills: students will be trained in acquiring a set of complex AI and programming skills in a restricted period of time. They will also be trained in applying those techniques on a given problem and report adequately the results.

Course Content

Reinforcement Learning is a set of techniques that can be used to solve sequential decision-making tasks. When combined with deep learning as a function approximator, these algorithms are referred to as deep RL and they

have been able to solve a variety of tasks that were previously out of reach for a machine (e.g. the game of Go). Thus, deep RL opens up many new applications in domains such as healthcare, robotics, smart grids, finance, and many more.

This course will provide an introduction to how real-world problems can be formalized as sequential decision-making problems (MDP, POMDPs, meta RL) as well as how (deep) reinforcement learning techniques such as deep Q-networks and actor-critic methods can be useful. The course will provide the theoretical foundations as well as an introduction to the practical skills necessary to apply (deep) RL techniques. Students will also develop their coding skills by developing (deep) RL agents.

Additional Information Teaching Methods

The course will be centered on the practical task of designing intelligent agents that can solve a sequential decision-making problem close to real-life (e.g. optimizing the use of a damp by storing/releasing energy on the electricity grid, given some past time series of the daily electricity price).

For the first week, there will be the equivalent of 4 hours of online pre-recorded lectures, one/two practical sessions (1h30 each), one or two Q&A sessions in a flipped classroom style (1h30 each). All students should know the basic elements of RL and deep learning by the end of the first week.

During the second week, there will be 4 lectures (each 1h30) covering specific topics of (deep) reinforcement learning as well as one/two practical session(s) and a test for the theory part.

The last two weeks will focus on implementing RL algorithms to solve the project and write a scientific report describing the results obtained. This part also encompasses a significant amount of self-study to familiarise oneself with the (deep) RL algorithms and their implementations.

Method of Assessment

The final grade takes the following elements into account:

Participation to the group work (seminar group) sessions via the attendance sheet (10% of the grade).
Test (10% of the grade)
|Intermediate presentation (10% of the grade)
Report+code and final presentation (70% of the grade)

A resit for the report+code and final presentation (70% of the grade) is possible but there is no resit possible for the other parts. An individual component in the grade is included in the final presentation and should be passed by each student in order to pass the course.

Literature

- Sutton, Richard S., and Andrew G. Barto. Introduction to reinforcement learning. Vol. 135. Cambridge: MIT press, 1998. Recommended, but not compulsory. There will be a reader.
- François-Lavet, Vincent, et al. "An Introduction to Deep Reinforcement Learning." Foundations and Trends® in Machine Learning 11.3-4 (2018): 219-354.

Recommended background knowledge

The minimal required background is basic machine learning concepts (supervised learning, the concept of loss functions, etc.). Some notions about Markov Decision Processes (MDPs) and practical knowledge of deep learning are also strongly recommended.

Quantitative Financial Risk Management

Course Code	E_FIN_QFRM
Credits	6
Period	P5
Course Level	400
Language Of Tuition	English
Faculty	School of Business and Economics
Course Coordinator	dr. C.S. Bos
Examiner	dr. C.S. Bos
Teaching Staff	dr. C.S. Bos
Teaching method(s)	Seminar, Computer lab, Lecture

Course Objective

Deep understanding and ability to implement modern quantitative risk measurement and management techniques, in the areas of market, credit and model risk.

Course Content

The course focuses on financial risks facing financial institutions, such as market, credit and model risks.

The course will encompass both theoretical and applied aspects of risk management. This course will give you a solid fundamental for measurement and management of financial risks, knowledge of newest quantitative methods and the ability to apply your knowledge in financial environment. The lectures are complemented by practical assignments designed to maximally match actual risk management applications in banking environment. For this course you need a strong quantitative focus and affiliation with statistics and probability as well as (some) affiliation with finance, as well as solid programming experience.

Additional Information Teaching Methods

Lectures and practice sessions.

Method of Assessment

Assignments.

Entry Requirements

Course on Investments or another, similar Finance-related course is necessary to follow this course. Solid background in advanced statistics, probability and programming is necessary.

Literature

Embrechts, Frey and McNeal "Quantitative Risk Management". Lecture notes and slides.

Additional Information Target Audience

Honours programs in Quantitative Risk Management, Finance and Technology. Students from other master programs (such as Business Analytics, Mathematics, Econometrics) are also welcome, as long as they comply with entry requirements.

Recommended background knowledge

Solid background in statistics and probability theory, advanced implementation skills (R, Python or any other computer package) Knowledge of main concepts in the areas of investments and derivatives.

Quantitative Investing

Course Code	E_FIN_QI
Credits	6
Period	P1
Course Level	400
Language Of Tuition	English
Faculty	School of Business and Economics
Course Coordinator	M.A. de Vries
Examiner	M.A. de Vries
Teaching Staff	N.M. Bonnmann, M.A. de Vries
Teaching method(s)	Study Group, Lecture

Course Objective

This course aims to deepen your knowledge in the field of quantitative investments.

After completion of the course, you should:

- Be familiar with the main quantitative investment strategies

- Understand investor behavior that influences financial markets.
- Be able to do advanced asset pricing tests on large data sets
- Be able to develop and implement quantitative investment strategies

Course Content

The field of quantitative investments consists of building investment strategies that are based on quantitative analyses. The course starts with a recap of the traditional portfolio theory and the Capital Asset Pricing Model (CAPM), with an emphasis on their practical implementation. Subsequently, current quantitative investment strategies will be discussed and implemented. Finally, the course goes into the details of investor behavior and provides explanations for the behavior that is observed. Understanding investor behavior is essential for developing new (quantitative) investment strategies.

Additional Information Teaching Methods

Lectures and tutorials. Lectures will have a stronger focus on theory and students will work on empirical applications in the assignment(s).

Method of Assessment

Assignment(s) and exam

Entry Requirements

- Investments (Bodie, Kane, and Marcus)
- Statistics (intermediate level)

You are not allowed to take this course if you have already taken Advanced Asset Pricing

Literature

Selected articles.

Additional Information Target Audience

- Duisenberg Honours Program in Quantitative Risk Management
- Financial Econometrics
- Business Analytics

Queueing Theory

Course Code	XMM_400397
Credits	6
Period	P4+5+6
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. S. Bhulai
Examiner	
Teaching Staff	
Teaching method(s)	

Course Objective

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Course Content

This course is part of the joint national master programme in mathematics. For schedules, course locations and course descriptions see https://elo.mastermath.nl

Additional Information Teaching Methods

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Method of Assessment

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Additional Information Target Audience

mMath

Custom Course Registration

You have to register your participation in each Mastermath course via https://elo.mastermath.nl/login/. Registration is mandatory and absolutely necessary for transferring your grades from Mastermath to the administration of your university.

Research Seminar Business Analytics

Course Code	XM_0054
Credits	6
Period	P4+5
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. A. Zocca
Examiner	dr. A. Zocca
Teaching Staff	dr. J. Berkhout, drs. J. Slootweg, dr. A. Zocca, dr. S.L. van der Pas
Teaching method(s)	Lecture

Course Objective

After taking this course, the students will be able to:

- Read and study a scientific paper in depth so that its main idea(s) and all its arguments and methods can be explained.
- Identify the relevant scientific literature around a specific practical research question and carry out solid desk research based on scientific literature (a sort of "review") related to that question.
- Present and summarize academic papers, both orally and in writing.
- Critically discuss scientific literature.

Course Content

This course aims to prepare the student for screening, reading, and understanding academic literature. This is particularly useful in particular for the master project, where the student should consider relevant literature for the research question at hand. Getting acquainted with the literature consists of two elements: (i) being able to read and study in depth a specific academic paper, and (ii) carrying out a review (desk research) of scientific literature related to a chosen research question. Both of these elements are addressed during the seminar.

Additional Information Teaching Methods

Each student gives two presentations (each about a different academic paper) in a seminar format, getting feedback from the lecturer and the other students. Each student also writes a short report/literature review centered around the two selected academic papers and the underlying common research question. Each student is asked to provide constructive and helpful feedback on the presentations and reports of his/her peers. The course manual provides more details.

Method of Assessment

Presentations (2x), written final report, attendance, and quality of given feedback. More details may be found in the course manual.

Scheduling

Course Code	XMM_400396

Credits	6
Period	P4+5+6
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. S. Bhulai
Examiner	
Teaching Staff	
Teaching method(s)	

Course Objective

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Course Content

This course is part of the joint national master programme in mathematics. For schedules, course locations and course descriptions see https://elo.mastermath.nl

Additional Information Teaching Methods

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Method of Assessment

.

Additional Information Target Audience

mMath, mBA

Custom Course Registration

You have to register your participation in each Mastermath course via https://elo.mastermath.nl/login/. Registration is mandatory and absolutely necessary for transferring your grades from Mastermath to the administration of your university.

Statistical Models

Course Code	X_400418
Credits	6
Period	P1+2
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. E.N. Belitser
Examiner	dr. E.N. Belitser
Teaching Staff	dr. E.N. Belitser
Teaching method(s)	Seminar, Lecture

Course Objective

The goals of this course are to get acquainted with some of the most commonly used statistical models, to learn how to apply these models in valid settings, and to understand the basic theory behind these models.

Course Content

ANOVA models, nonlinear regression models, generalized linear models, time series (possibly some related topics).

Additional Information Teaching Methods

Lectures and tutorials.

Method of Assessment

Assignments and examination.

Literature

Lecture slides and lecture notes "Statistical Models" by M.C.M. de Gunst.

Additional Information Target Audience

MSc BA, MSc Mathematics

Additional Information

Students will use the statistical package R (www.r-project.org) for data analysis.

Explanation Canvas

All the information about the course (and group enrollment) will be available on Canvas.

Recommended background knowledge

Course(s) on statistics and probability (such as Statistics X 400004 and Probability Theory X_400622 or their equivalents) and a basic course on Linear Algebra are required; experience with statistical package R is strongly recommended.

Stochastic Processes for Finance

Course Code	X_400352
Credits	6
Period	P1+2
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. P.J. de Andrade Serra
Examiner	dr. P.J. de Andrade Serra
Teaching Staff	dr. P.J. de Andrade Serra
Teaching method(s)	Seminar, Lecture

Course Objective

After this course, the student will:

- be acquainted with the notion of pricing derivatives and arbitrage, and with the basics of discrete stochastic processes, including the concepts of martingales; be able to price derivatives in the binomial model.
- be familiar with the basics of continuous time stochastic processes, concepts of measure theory, Brownian motion, martingales; be able to verify whether the process is a Brownian motion and/or is a martingale.
- be familiar with the notions of stochastic integral Ito and SDE, the Ito's formula; be able to compute stochastic integrals, to use the Ito formula, and to derive properties of stochastic integrals and the solutions of SDEs.
- be familiar with the Black-Sholes model; be able to price options by the no-arbitrage and Girsanov's theorem, to compute the hedging portfolio and the Greeks, and to make connections to the corresponding Black-Scholes PDEs.
- be familiar with the extended Black-Sholes models; be able to price derivatives in those models, to make connection with the corresponding PDEs, and to apply the Black-Scholes machinery to the interest rates models.

Course Content

Financial institutions trade in risk, and it is therefore essential to measure and control such risks. Financial instruments such as options, swaps, forwards, etc. play an important role in risk management, and to handle them one needs to be able to price them. This course gives an introduction to the mathematical tools and theory behind risk management.

A "stochastic process" is a collection of random variables, indexed by a set T. In financial applications the elements of T model time, and T is the set of natural numbers (discrete time), or an interval in the positive real line (continuous time). "Martingales" are processes whose increments over an interval in the future have zero expectation given knowledge of the past history of the process. They play an important role in financial calculus, because the price of an option (on a stock or an interest rate) can be expressed as an expectation under a so called martingale measure. In this course we develop this theory in discrete and continuous time. Most models for financial processes in continuous time are based on a special Gaussian process, called Brownian motion. We discuss some properties of this process and introduce "stochastic integrals" with Brownian motion as the integrator. Financial processes can next be modelled as solutions to "stochastic differential equations". After developing these mathematical tools we turn to finance by applying the concepts and results to the pricing of derivative instruments. Foremost, we develop the theory of no-arbitrage pricing of derivatives, which are basic tools for risk management.

Additional Information Teaching Methods

Lectures (2 x 45min) and Tutorials (2x45). About 13 meetings of each are planned.

Method of Assessment

Two assignments covering computation and theory (up to 30% of final grade) and written examination/resit examination (at least 70% of final grade). There is no resit possibility for the assignments.

Literature

Lecture notes contain all of the material.
Additional, optional literature:

- Shreve, "Stochastic Calculus for Finance I: The Binomial Asset Pricing Model", Springer;
- Shreve, "Stochastic Calculus for Finance II: Continuous-time models", Springer.

Additional Information Target Audience

mBA, mBA-D, mMath, master Econometrics.

Additional Information

A significant part of the course is used to introduce mathematical subjects and techniques like Brownian motion, stochastic integration and Ito calculus. In view of this, the course is NOT meant for students who already followed the master course "Stochastic Integration" or "Stochastic differential equations". On the other hand, after completing this course, students may be motivated to follow other courses (like the two mentioned above) where stochastic calculus is treated in a deeper and more rigorous way.

Recommended background knowledge

Probability Theory (e.g., X_400622), Mathematical Analysis (e.g., XB_0009), Measure Theory (e.g., X_401028 or Mastermath course Measure Theoretic Probability)

Supply Chain Lab

Course Code	E_TSCM_SCL
Credits	6
Period	P4
Course Level	400
Language Of Tuition	English
Faculty	School of Business and Economics
Course Coordinator	dr. R. Paradiso
Examiner	dr. R. Paradiso
Teaching Staff	dr. R. Paradiso
Teaching method(s)	Study Group, Lecture

Course Objective

The course has the following five main learning objectives:

- Apply and test network optimization and integer programming techniques

- Formulate optimization models using an algebraic modelling language
- Solve integer and network optimization problems using dedicated software
- Analyze real-life industry problems, formulate and defend managerial recommendations
- Assess the capabilities and limitations of optimization models to solve real-life decision-making problems arising in distribution logistics

Course Content

Transport and distribution management often consists of interrelated decisions involving network design, production planning and scheduling, product storage and warehousing, transport and distribution. This course introduces students to these specific decision problems, and also equips them with tools and methods to solve these problems.

Mixed Integer Linear Programming (MILP) is a powerful tool for improving transport and distribution decision processes. As MILP is being used more and more often in the industry, the course wants to offer students key insights and sufficient training to be able to contribute in development and implementation projects in their professional career. To this end, dedicated software packages are introduced for modeling and solving distribution problems via MILP.

This course extends the scope of the Decision Making in Supply Chains course to more complex optimization problems arising in real-life transport and distribution planning. Solution approaches are illustrated by means of a selection of topics, e.g. transportation planning, facility location, network design, vehicle routing and scheduling, manpower planning, and rostering.

The teaching and learning are largely based on "learning by doing" with a number of cases in different industrial applications. A mixture of hearing lectures, tutorials, assignments and case studies offers students the best possible support to master sufficient skills to tackle real-life cases in distribution logistics.

Additional Information Teaching Methods

Lectures, Practical tutorials

Method of Assessment

Based on an individual exam and a course group project.

Time Series Models

Course Code	E_EORM_TSM
Credits	6
Period	P4
Course Level	400
Language Of Tuition	English
Faculty	School of Business and Economics
Course Coordinator	dr. I. van de Werve
Examiner	dr. I. van de Werve
Teaching Staff	dr. I. van de Werve, dr. K. Moussa
Teaching method(s)	Seminar, Lecture

Course Objective

This course provides students with the tools and insights necessary to carry out advanced time series analysis using state space methods. State space models are widely used in economics, finance, and policy analysis—for example, in forecasting inflation, analyzing financial volatility, or assessing climate measures. This course provides the foundation needed to apply these models in academic research, central banking, or data-driven policy making. By the end of the course, students will be able to independently design, implement, and interpret a complete econometric analysis using these models, both from a theoretical and empirical perspective.

The course focuses on building a solid theoretical foundation for time series analysis using state space methods. Students will explore the key principles underlying Kalman filtering techniques and their extensions, gaining a deep understanding of the methodologies used to model and forecast dynamic systems. The course also includes practical applications, where students will gain experience working with real data and develop the ability to clearly communicate complex theoretical results.

Course Content

The course integrates three key dimensions of time series analysis: theory, methodology, and computation—with state space models serving as the unifying framework.

In the first part of the course, the focus is on linear Gaussian models, with key methodologies including the Kalman filter, Kalman smoother, maximum likelihood estimation, and simulation smoothing. In the second part, the course shifts to nonlinear and non-Gaussian models, introducing advanced techniques such as the extended Kalman filter and smoother, mode estimation, importance sampling, and particle filtering. Finally, the course addresses computational challenges in working with multivariate time series.

Throughout the course, theoretical models will be illustrated using data from various domains, including finance, climate, and criminology. Students will also gain practical experience in implementing algorithms using real-world data.

The mathematical derivations rely on essential principles from econometrics, probability theory, linear algebra, statistics, and mathematical analysis. While theoretical rigor is emphasized, equal importance is placed on interpretation and practical application of the methods.

Additional Information Teaching Methods

The course structure ensures that students have multiple ways to engage with the material. Each week, the following is offered:

- Lectures: Introduce and explain the core concepts, models, and methods;
- Work group session: Focus on theoretical exercises and deeper understanding of model derivations;
- Assignment office hours: Participation is optional, but students are encouraged to use these sessions to discuss questions;
- Canvas discussion boards: Provide additional support for questions related to the course organization, theory, work group exercises, programming assignments, and the (practice) exam.

There is also one guest lecture during the course period, delivered by an expert from academia or industry, offering practical insights into the application of state space models in ongoing research or real-world projects.

Method of Assessment

The assessment is based on two components: an individual written exam (covering the theoretical part of the course) and a group computer assignment (covering the empirical part of the course). Further details on the format, deadlines, and expectations for both components are shared during the first lecture and on Canvas. Peer evaluation is facilitated through FeedbackFruits – Group Member Evaluation, which allows group members to reflect on team contributions and resolve any disputes transparently.

Entry Requirements

This is an advanced course. Required background includes: probability theory, linear algebra, statistics, mathematical analysis, and programming, as well as their application in time series econometrics. To participate, students are expected to have followed the courses offered in periods 1, 2, and 3 of the MSc Econometrics programme. Assignments involve coding in, for example, Python or Matlab. Refresher materials on key prerequisites will be available on Canvas before the course begins.

Literature

The primary textbook for the course is Durbin, J., & Koopman, S.J. (2012). Time Series Analysis by State Space Methods, Second Edition. Supporting material is provided through Canvas. A detailed list of the study material is available on Canvas before the start of the course.

Explanation Canvas

During the course period, Canvas is used for all official communication, distributing lecture slides and exercises (with solutions), registering for office hours, and submitting assignments. Students are expected to monitor Canvas regularly for updates and announcements.